

Power Platform Bootcamp

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Cloud Solution Architect – Business Applications
Microsoft

Agenda

Time	Activity Title
0930	Kick Off
0940	Power Platform Overview
1000	Power Apps Overview
1030	Break
1045	Power Apps Screens, Controls and Formulas
1130	Lunch
1300	Hands on Exercises
1500	Break
1515	Hands on Exercises
1700	End

Session Information

Target Audience

New to Power Platform and creating applications. Basic knowledge of Microsoft 365 applications such as Microsoft PowerPoint, Microsoft Teams & Microsoft Excel.

Objectives

- Gain a foundational understanding of Microsoft Power Platform, including the core components
 - Power Apps
 - Power Automate
 - Power BI
- Learn how to create simple, low-code applications, automate processes, and analyse data using built-in tools.
- Attempt hands-on exercises and explore how these tools can be used to solve real-world business challenges and drive productivity, while gaining confidence in building scalable solutions without extensive programming knowledge.

Power Platform Overview

The background of the slide is a gradient of blue, transitioning from a darker shade at the top to a lighter shade at the bottom. On the right side, there are several curved, glowing white lines that sweep across the frame. These lines are composed of many small, bright white dots, giving them a sense of motion and energy. The overall aesthetic is modern and technological.

Microsoft Power Platform

Power Platform is a rapid application development platform that helps citizen developers with limited coding abilities, to innovate and build new solutions faster in an environment and user experience that feels like PowerPoint and Excel, called a low-code environment.



Power Apps
Application development



Power Automate
Process automation



Copilot Studio
Intelligent chatbots



Power Pages
Website



Power BI
Business analytics



**Data
connectors**



AI Builder



**Microsoft
Dataverse**



Power Fx

Why Low Code Development?

750 million

new apps need to be built in by 2025¹

75% of apps to be built on Low Code⁴

Power Platform

33 million

monthly active users²

25%

reduction in time to complete tasks³

Used by **97%** of Fortune 500



Traditional Software is slow and expensive



Pursuing Digital Transformation



Security & Compliance Risks

1. IDC, Connected Development: Delivering Digital-First Excellence. October, 2022.
2. Gartner, Forecast Analysis: Low Code development Technologies. January, 2021.
3. Microsoft Fiscal Year 2023 Third Quarter Earnings Conference Call. April, 2023.
4. Gartner, Forecast Analysis: Low-Code Development Technologies. October 2022

A Platform For All Makers

Citizen Developer, IT Administrator, Professional Developer, App Wizard. We've got you



Intuitive to use & easy to learn for Citizen Developers

Build apps fast with a point-and-click approach to app design. Choose from a large selection of templates or start from a blank canvas

Easily connect your app to data and use Excel-like expressions to easily add logic

One admin center to rule them all for IT Administrators

No compromises on governance and security. One centralized view and management of all your 1st & 3rd party apps, eliminating shadow IT

GDPR compliance, and enterprise grade security, consistent with the experience across O365 & D365

World class Pro Developer support and ALM

Experience the full range of development and ALM functionality with rich pro developer tools like Visual Studio and DevOps

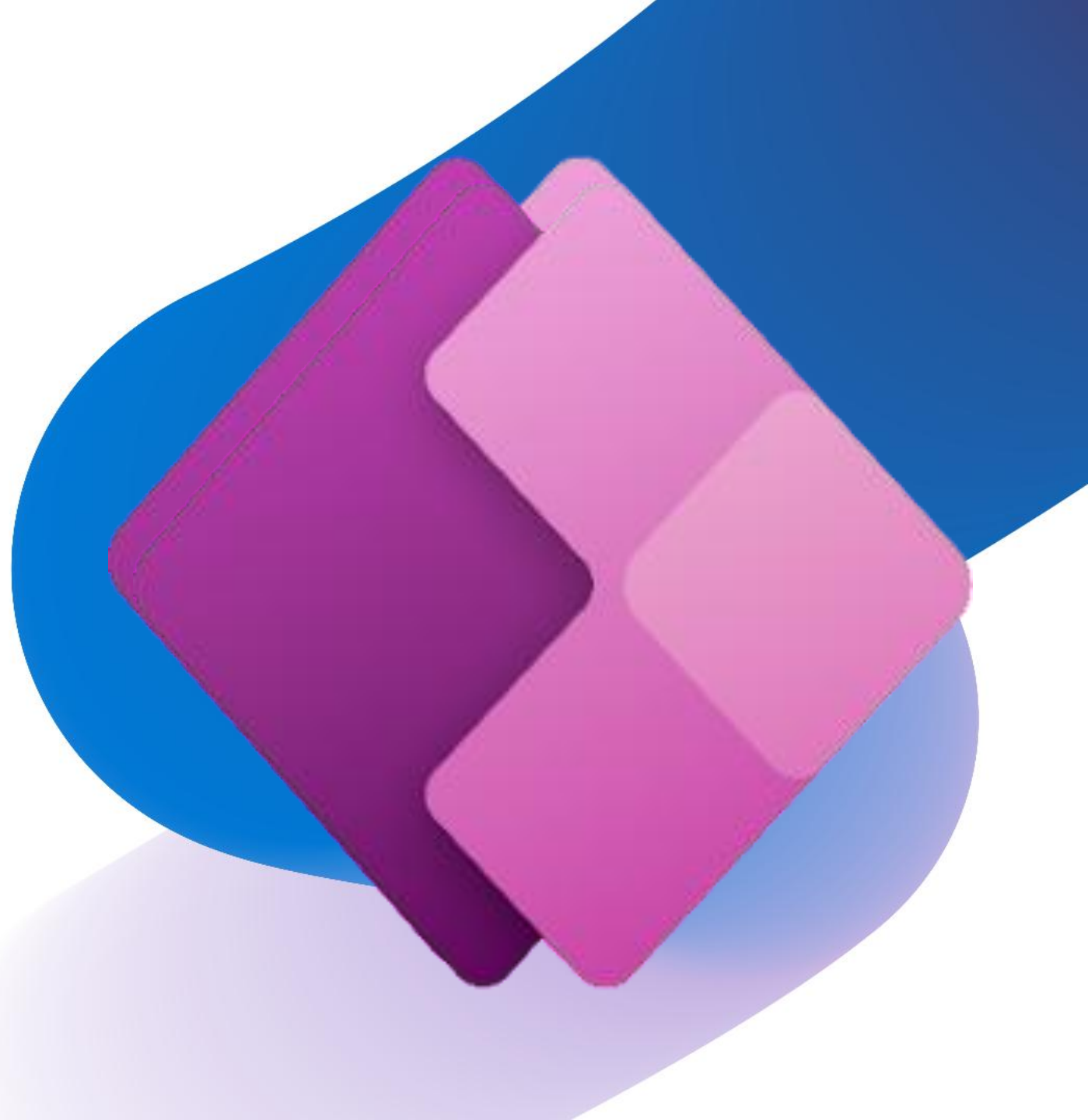
Reuse your current IP and skills – Power Apps component framework, CLI and VS Code create instant value for the business



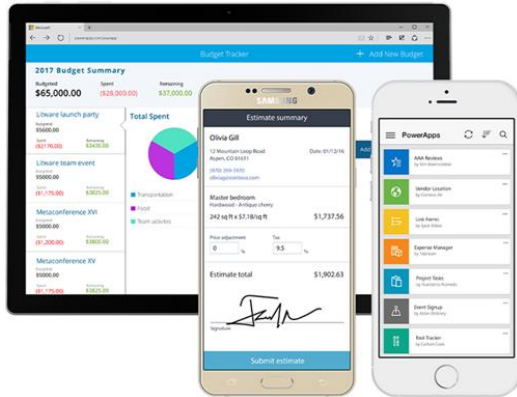
Power Apps Overview

What are Power Apps?

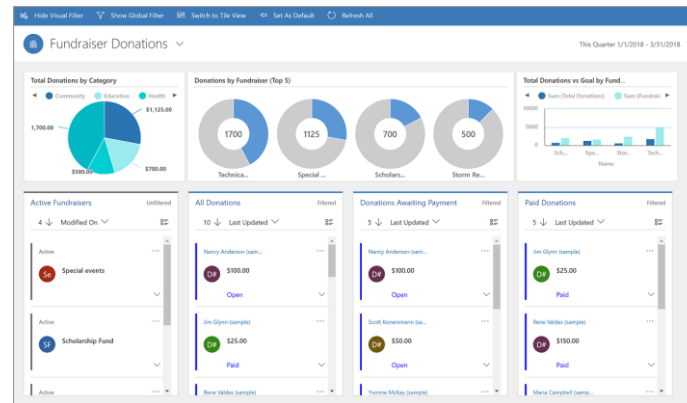
- Applications that anyone can build for the web and mobile devices—
No coding experience.
- A **drag and drop** interface allows you to add and configure what you need to solve the problem at hand with controls like buttons and galleries.
- An **Excel-like formula** bar allows you to dig in a little deeper and apply more complex rules and logic. See results in real time.
- You can **extend Power** Apps by creating customized connections and building blocks that can be used across multiple programs
- Use the App Via Microsoft Teams, on your phone or the on web



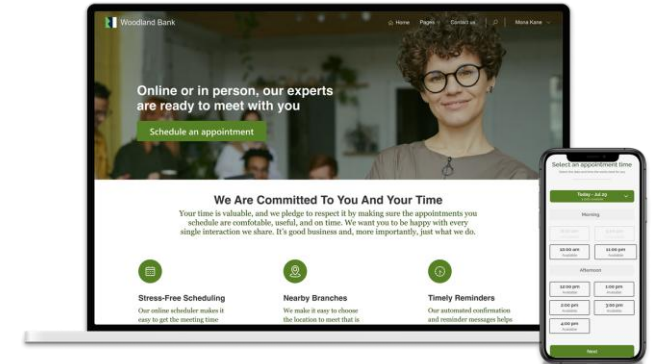
Types of Power Apps



Canvas apps start with user experience, with pixel-perfect control to build highly customized task- and role-based apps that mash up data from 700+ sources



Model-driven apps leverage your data model, relationships, and business processes to automatically generate immersive, responsive applications



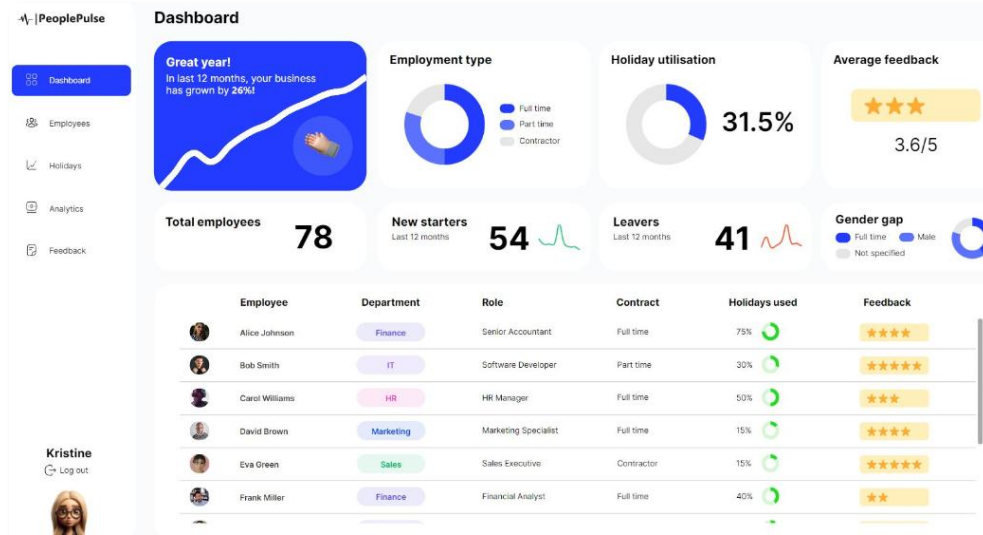
Power Pages an easier way to create modern, secure, responsive business websites to rapidly engage your customers, partners and communities



The focus should be on Canvas Apps today as the others tend to more complex. It is sufficient to understand the use case for the others.

Power Apps Examples

Use apps as an entry point to drive automation or visualize information



Build modern applications that can aid business decision making

FABRIKAM RESIDENCES MOVE OUT INSPECTION

Residence Info

Resident Name: [Text Box]
Apt #: [Text Box]
Floor Plan: 2bed, 2 bath

Inspection Type Moving out
Moving Out Date 04/09/2018
Inspected By [Text Box]
Inspection Date 04/11/2018

Inspection Checklist

Items	Passed	Repair	Replace	Notes
Flooring	☒	☐	☐	
Appliance	☒	☐	☐	
Ceiling	☒	☐	☐	

Top 9 Repairs

Repair needed? ☒ YES
Cleaning crew? ☒ YES

☒ Kitchen ☐ Bathroom ☒ Master Bedroom ☒ Living Room

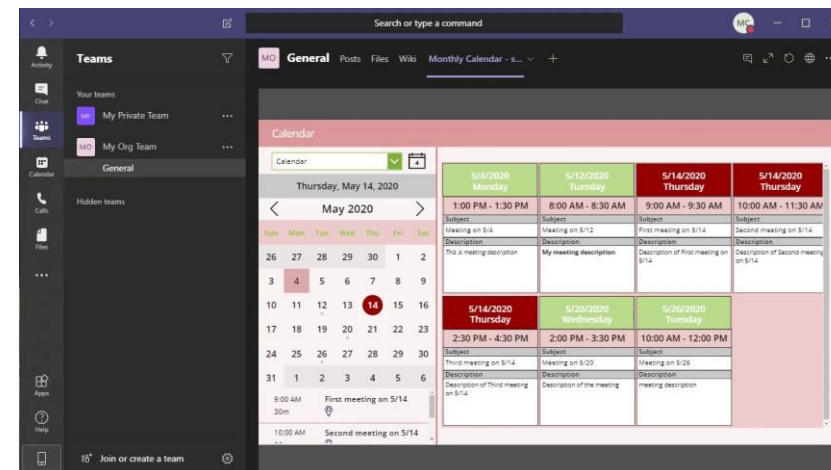
Submit **Cancel**

Create simple mobile ready applications

Allocations

Search items

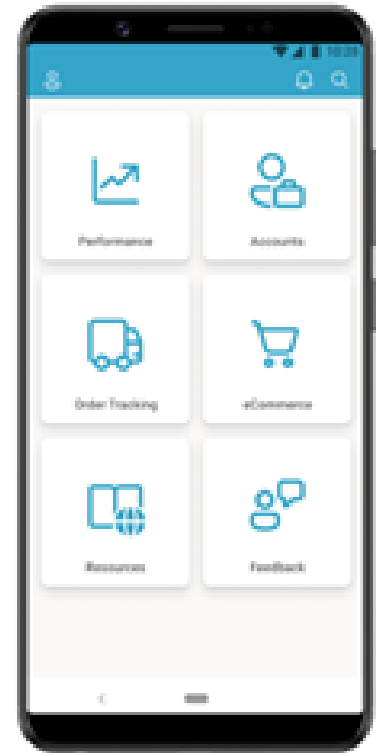
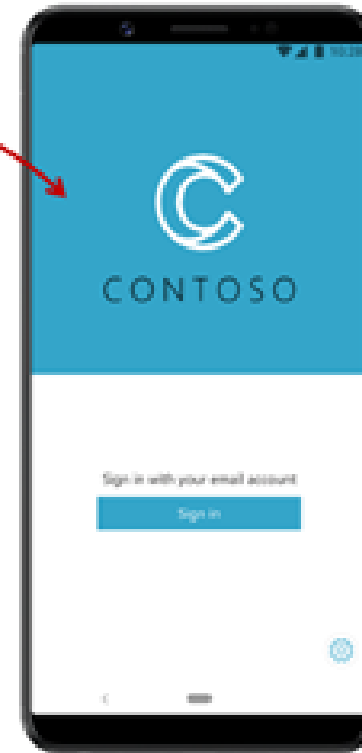
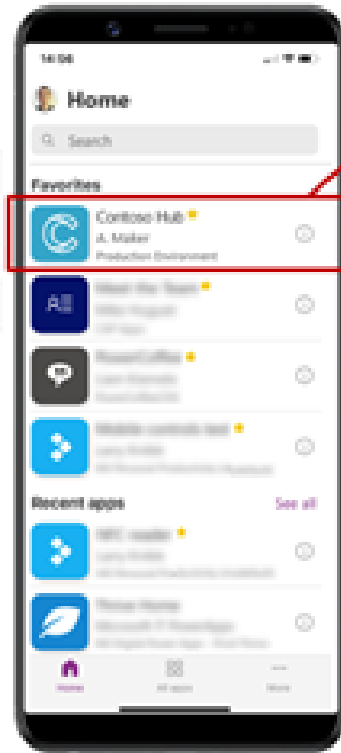
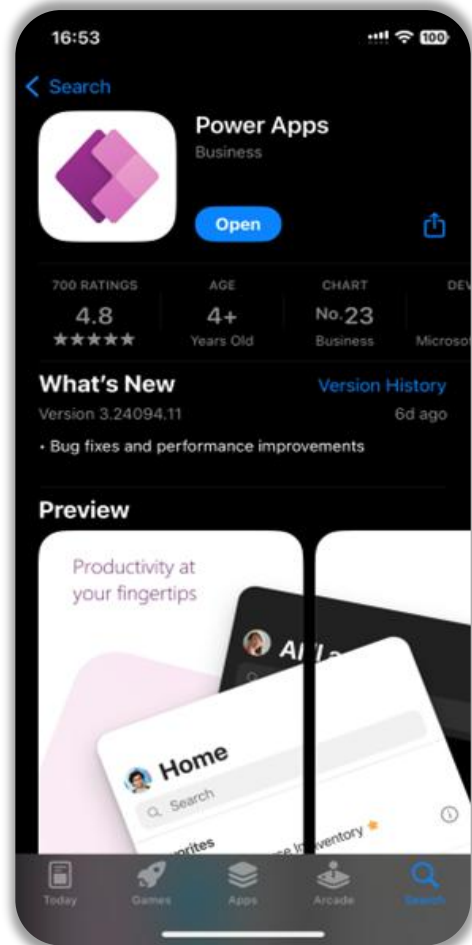
Display item 1	Project A	>
Display item 0.5	Project B	>
Display item 0.5	Project B	>
Display item 1	Project K	>



Access apps from a wide variety of sources

Power App mobile access

Access multiple Power Apps via the "Power Apps" App

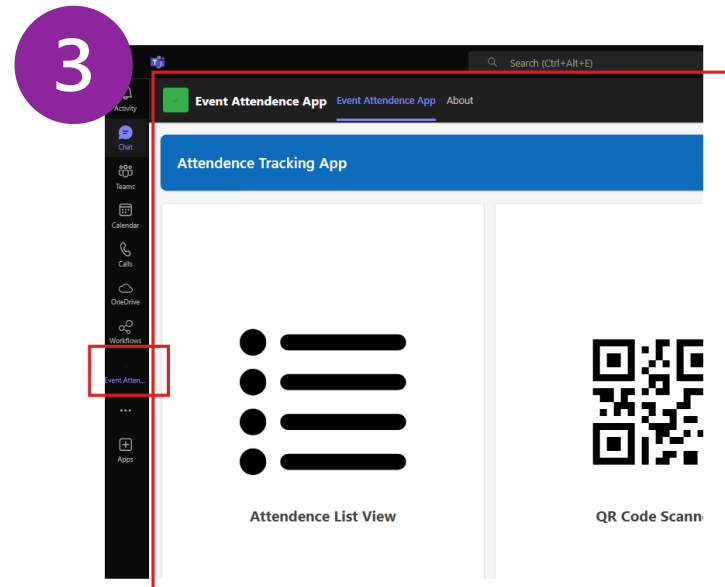
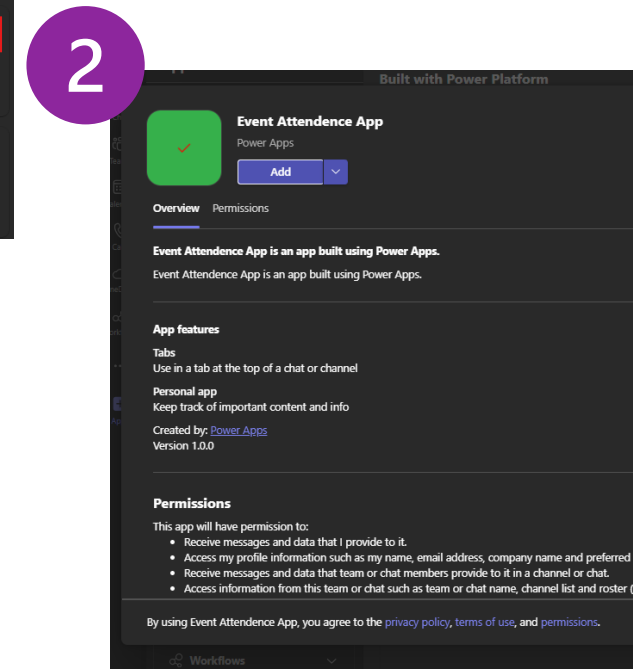
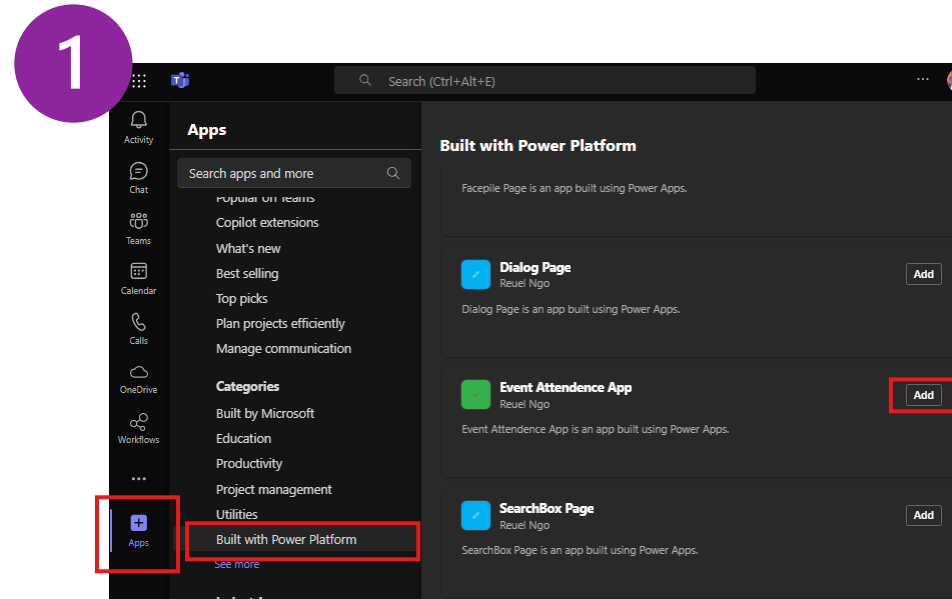


Accessing Power Apps in Teams – Pinned on Sidebar

Add a Power App to the side bar for easy access in Microsoft teams

1. Search for the Published App in Teams
2. Click on Add
3. Access it directly from Microsoft Teams.

With this, users can quickly access essential apps without leaving Teams. This is good for frequently used apps.

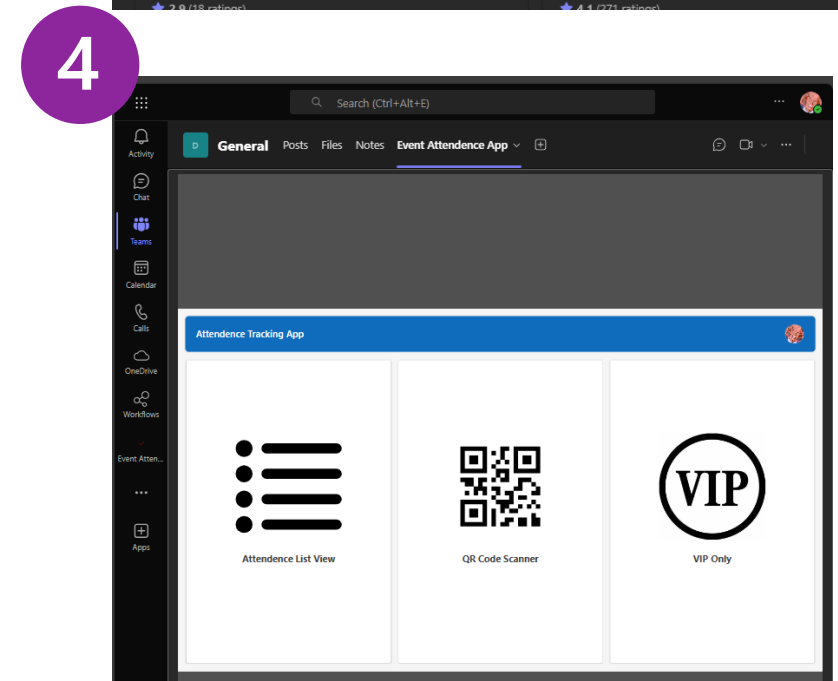
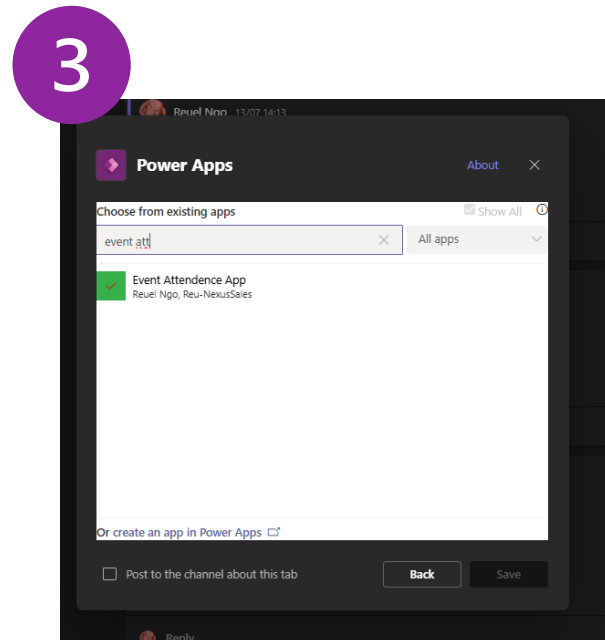
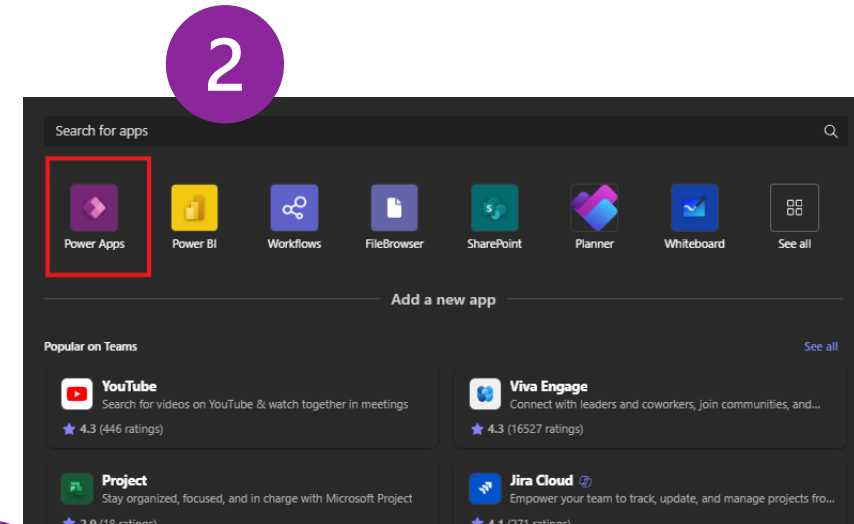
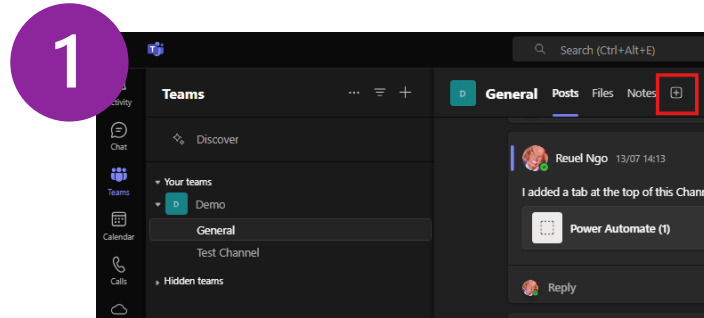


Accessing Power Apps in Teams – Pinned on a Channel

Add a Power App to a channel for easy group access in Microsoft teams

1. Navigate to your channel of choice and click on add.
2. Click on Power Apps
3. Search for the App you wish to add
4. Use the App

When Apps are critical for team collaboration, pinning an app in a channel allows users to use relevant apps without leaving the channel.





Power Apps

Connectors

Types of connectors

- Prebuilt Connectors – ready-made connectors that allows you to connect to apps and data in the cloud
 - Standard Connectors – Requires a Power Apps Seeded License
 - Premium Connectors – Requires a Power Apps Premium License
- Custom Connectors – Connectors that you build to support your apps or data that might not be residing in the cloud



Popular Standard connectors



Excel Online
(OneDrive)



Office 365
Outlook



OneDrive for
Business



Microsoft Teams



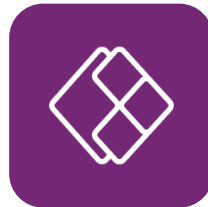
Microsoft To-Do
(Business)



Office 365
Users



Microsoft Forms



Power Apps
Notification V2



SharePoint



Planner

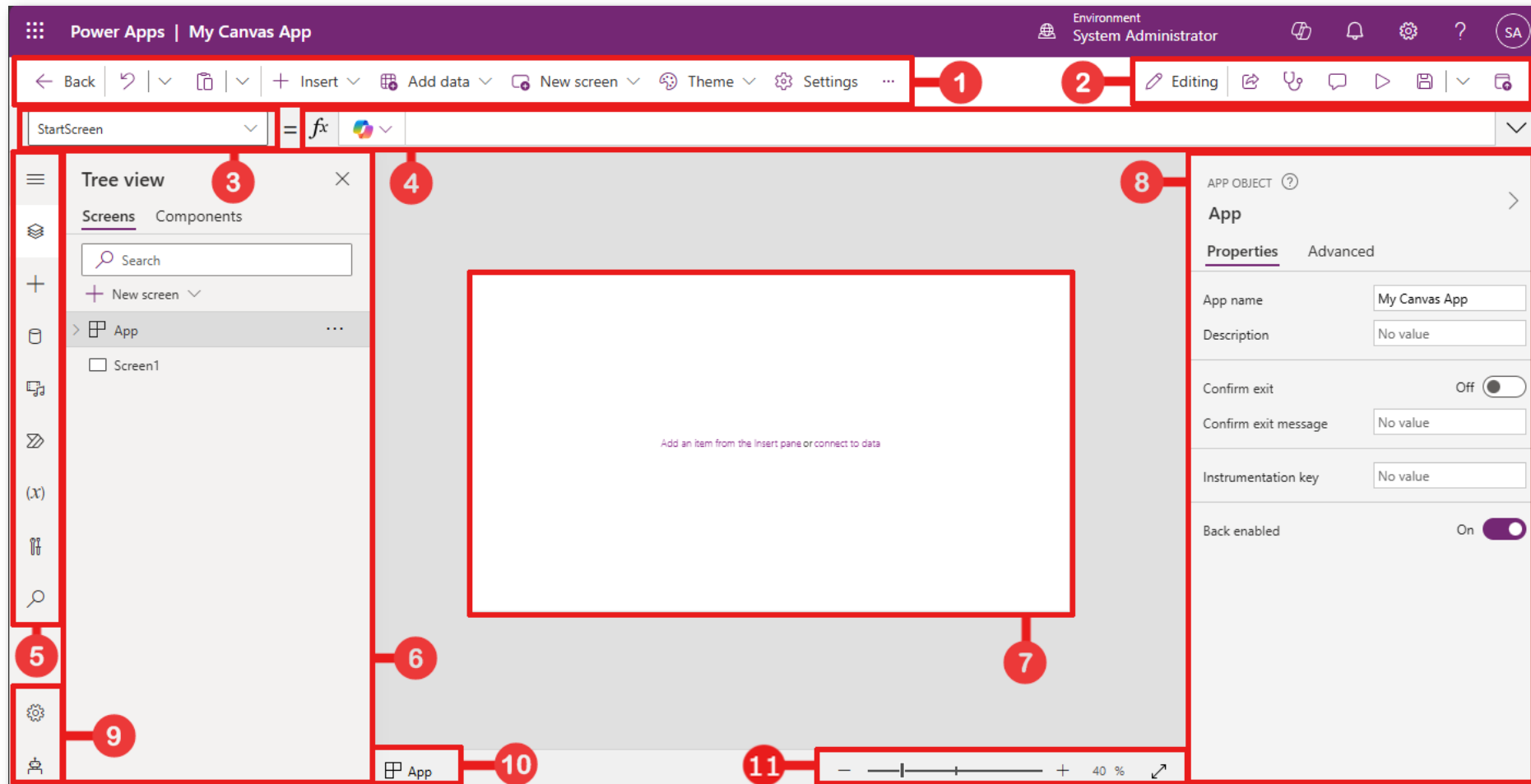


Power Apps

Screens and Controls

Power Apps Studio: The Designer's Toolbox

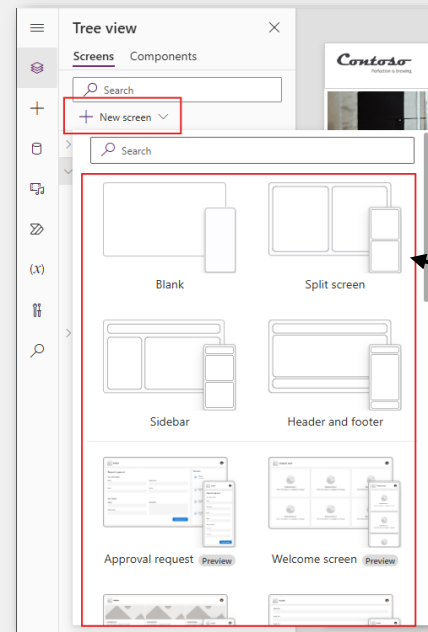
You can use Power Apps Studio to design, build, and manage your canvas app.



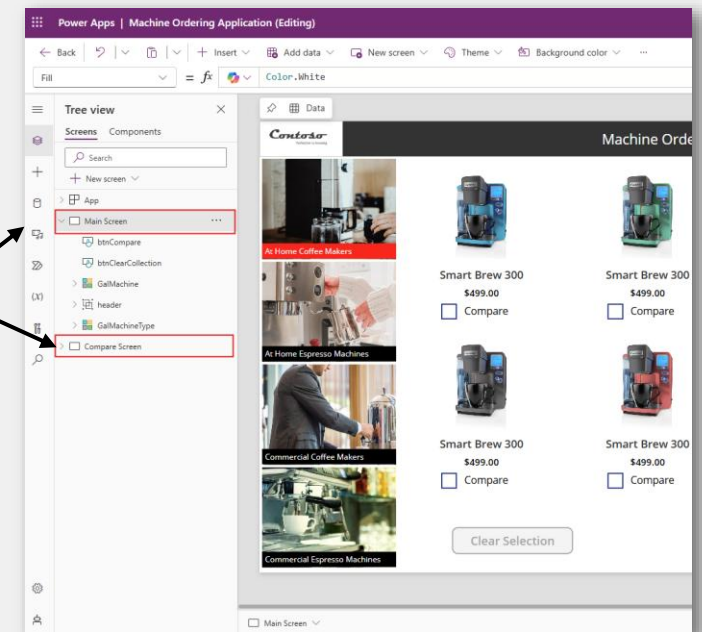
1. Power Apps Studio modern command bar
2. App actions
3. Properties List
4. Formula bar
5. App authoring menu
6. App authoring options
7. Canvas/screen
8. Properties pane
9. Settings and virtual agent
10. Screen selector
11. Change canvas screen size

Screens and controls

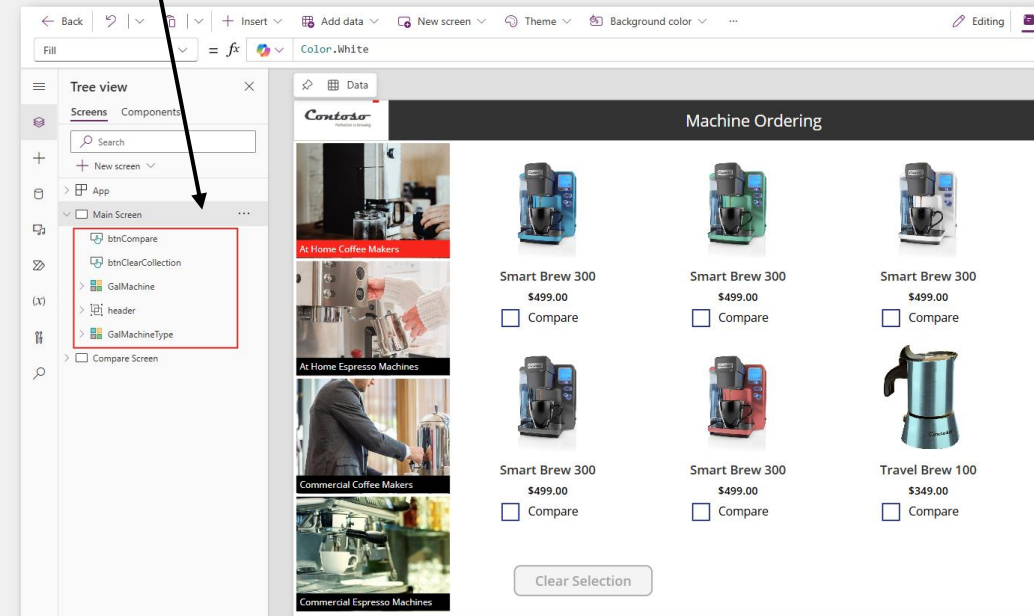
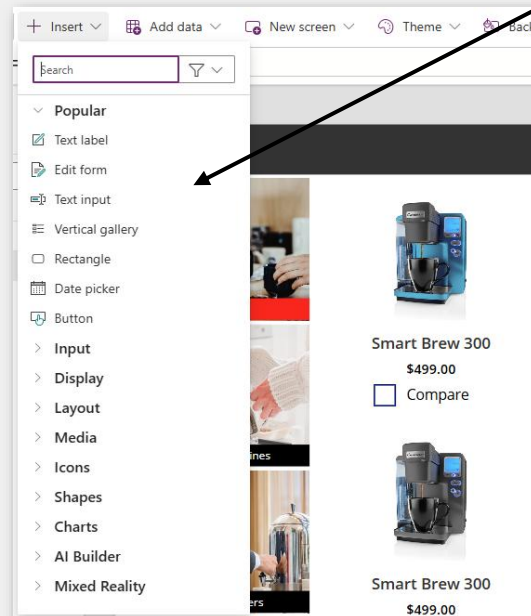
- Screens – Most apps have multiple Screen controls that contain Label controls, Button controls, and other controls that show data and support navigation.
- Controls – UI elements that you can control the properties of in your app.



Screens

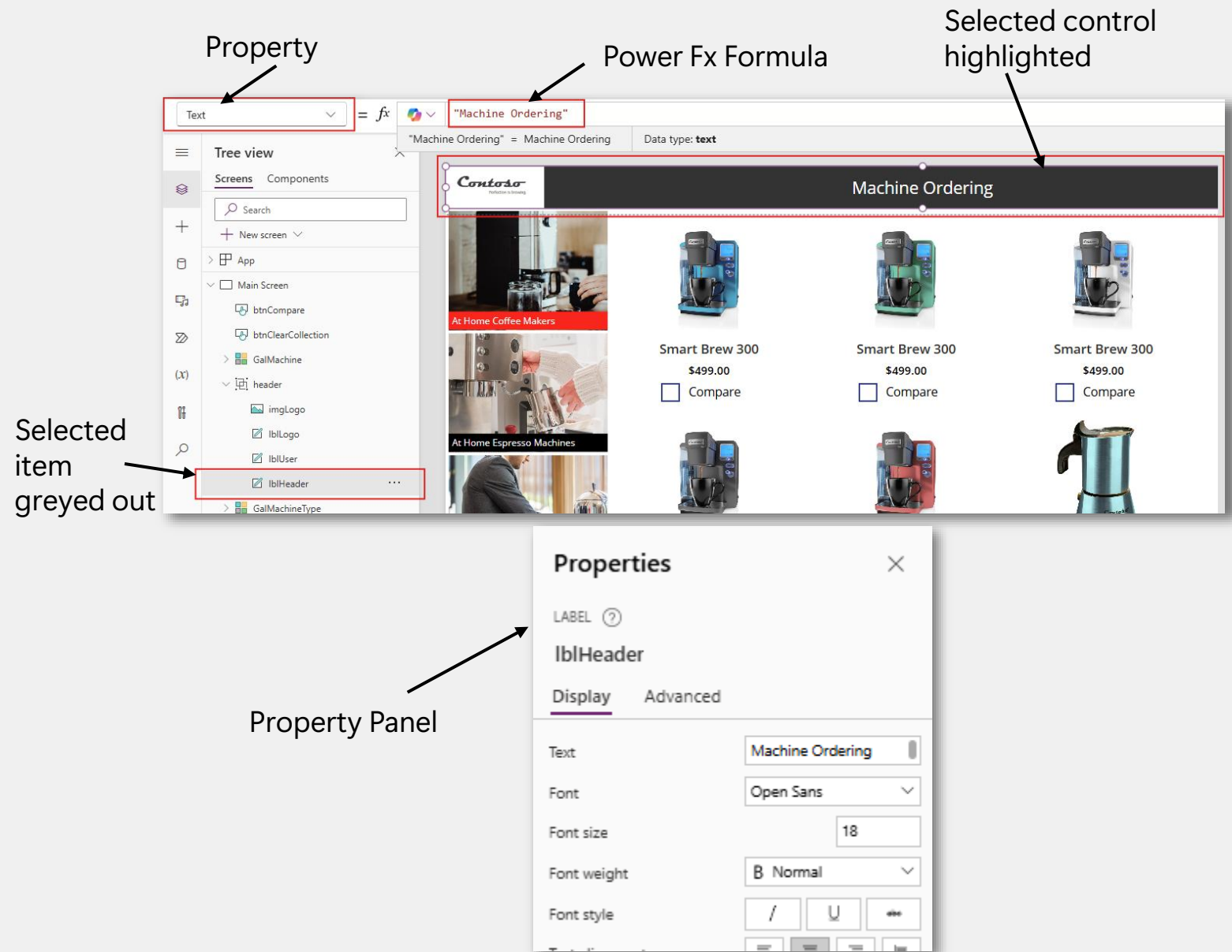


Controls



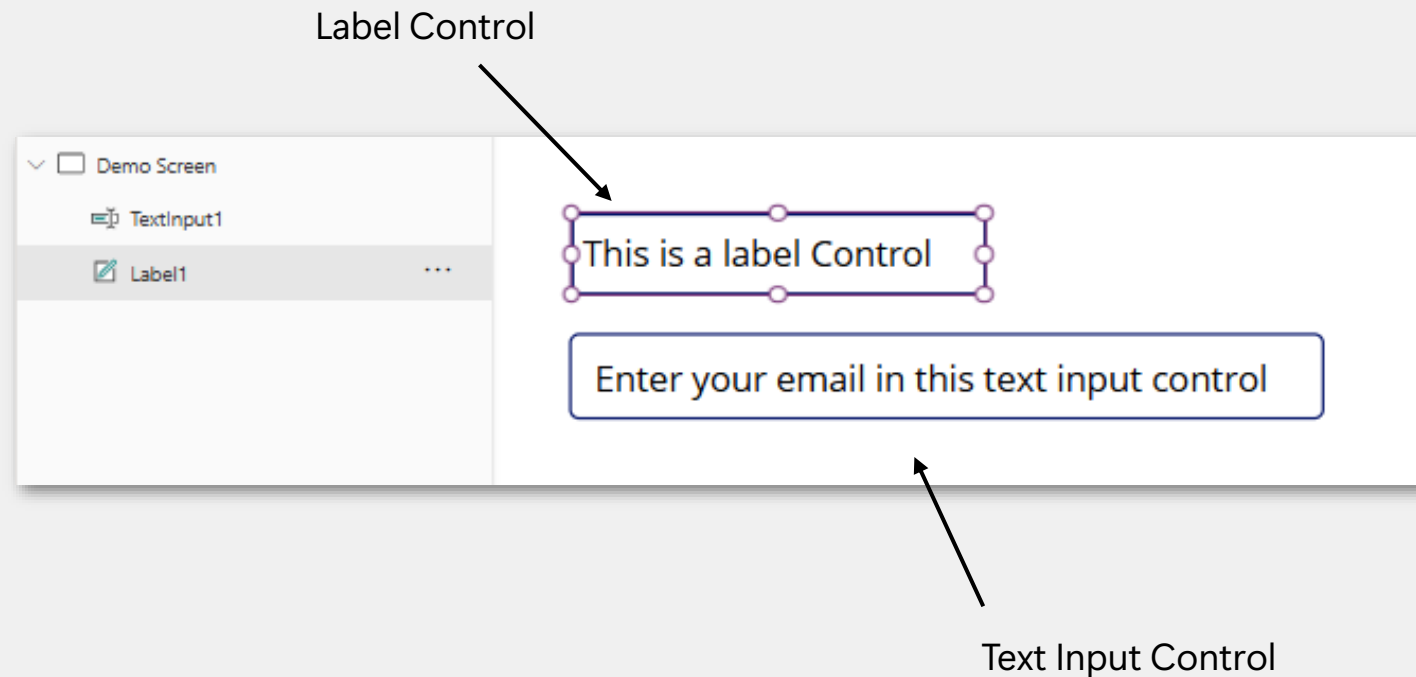
Screens and controls – Continued

- Screens – Most apps have multiple Screen controls that contain Label controls, Button controls, and other controls that show data and support navigation.
- Controls – UI elements that you can control the properties of in your app.



Text input and Label Controls

- Label Control – Shows data such as text, numbers, dates, or currency.
- Text input Control – A box in which the user can type text, numbers, and other data.



Forms

- Display Form Control – A user can display all fields of a record or only the fields that you specify
- Edit Form Control – A user can edit those fields, create a record, and save those changes to a data source

Edit Form Control

The screenshot shows the 'Edit Form Control' interface. The left pane displays a tree view with the following structure:

- Compare Screen
- Demo Screen
 - Button1
 - Form1
 - Address 1: City_DataCard2
 - StarVisible3
 - ErrorMessages3
 - DataCardValue11
 - DataCardKey11
 - Credit Limit_DataCard2
 - Email_DataCard1
 - Industry_DataCard2
 - Last On Hold Time_DataCard1
 - Account Name_DataCard2
 - Number of Employees_DataCard2
 - Preferred Method of Contact_DataCard2
 - Annual Revenue_DataCard2

The main area shows a form with the following fields:

- * Account Name (text input)
- Address 1: City (text input)
- Credit Limit (text input)
- Email (text input)
- Industry (dropdown menu)
- Last On Hold Time (date and time picker)
- Number of Employees (text input)
- Preferred Method of Contact (dropdown menu)
- Annual Revenue (text input)

A 'Submit' button is located at the bottom right of the form. An arrow points from the 'Edit Form Control' caption to the 'Fields' tab in the top right of the form area.

The screenshot shows the 'Display Form Control' interface. The left pane displays a tree view with the following structure:

- FormViewer1
 - Birthday_DataCard1
 - Email_DataCard2
 - Marital Status_DataCard1
 - Full Name_DataCard1
 - Job Title_DataCard1
 - Mobile Phone_DataCard1

The main area shows a form displaying data for a record:

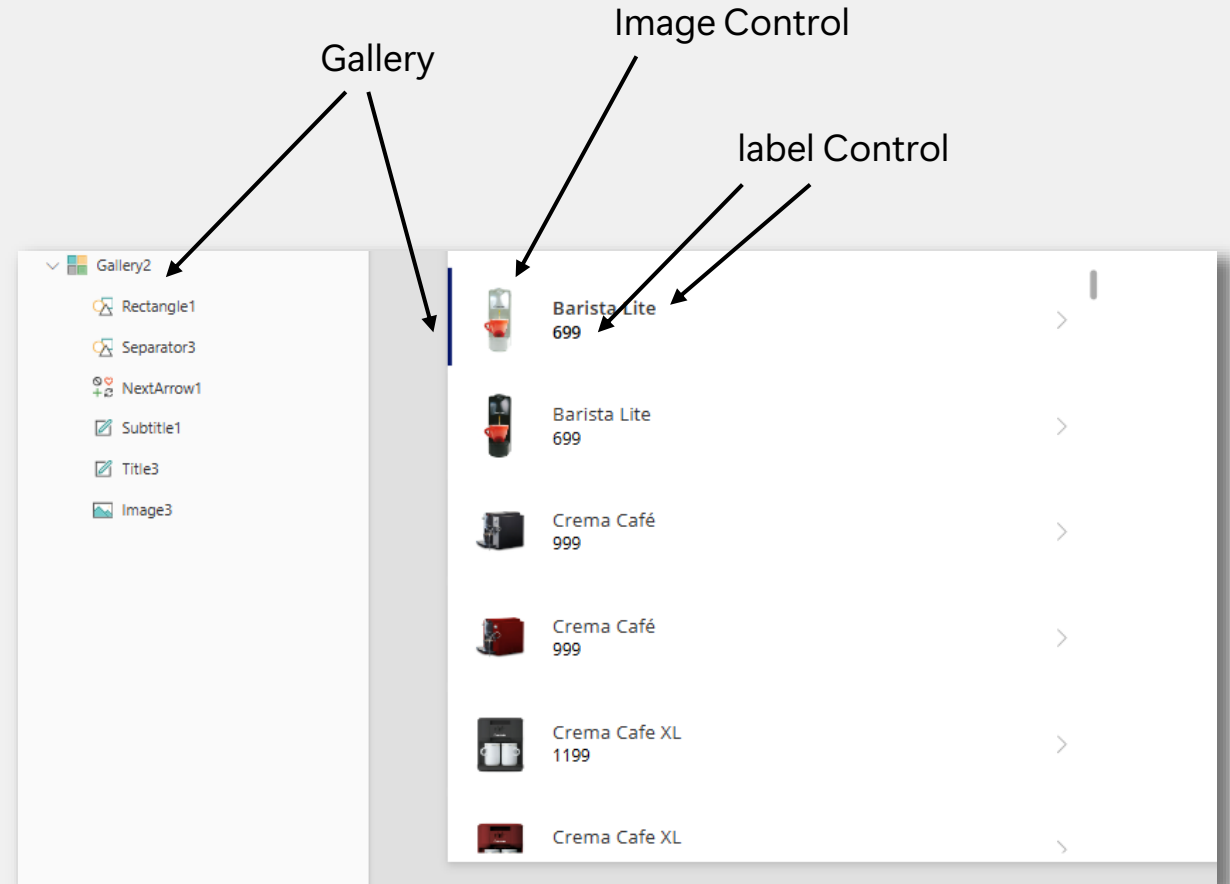
- Full Name: Mark Tan
- Email: marktan@contoso.com
- Birthday: 09/02/1988
- Marital Status: Married
- Job Title: CEO
- Mobile Phone: (empty)
- Preferred Method of Contact: Any
- Website: <https://contoso.com>

An arrow points from the 'Display Form Control' caption to the bottom of the form area.

Display Form Control

Galleries

- A Gallery control can show multiple records from a data source, and each record can contain multiple types of data.
- Each data field appears in a separate control which you can configure within the Gallery control.
- The template appears as the first item inside the gallery:
 - On the left edge of a Gallery control in horizontal/landscape orientation.
 - And at the top of a Gallery control in vertical/portrait orientation.





Power Apps

Formulas

What are functions in Power FX?

A function is like giving instructions to an app in a specific order to get a certain result. Just like how you would follow steps when doing something in real life, a function expects you to:

- Give the right inputs: These are the things you need to get the result you want (like the options in an order).
- Follow a specific format: You have to provide the inputs in the correct sequence so the app knows what you're asking for.
- Receive an expected outcome: Based on the inputs you provide, the function will give you a result.

For example, if you order a bubble tea, you can't just say random words like "sugar pearls ice tea," the cashier would be confused. You need to give instructions in the right order:

- What flavour do you want?
- How much sugar?
- How much ice?
- What toppings?

If we were to model ordering bubble tea like a function, it would look something like this:

OrderBubbleTea(flavor, sugarLevel, iceLevel, toppings)

- **flavor** is first because that's the main part of your order.
- **sugarLevel** and **iceLevel** are next, because they modify how your drink is made.
- **Toppings** come last, and they are **optional additions**.

Function Examples

```
OrderBubbleTea( "classic milk tea", "50%", "regular ice", "pearls" )  
OrderBubbleTea( "green tea", "25%", "less ice")
```

Function Executed

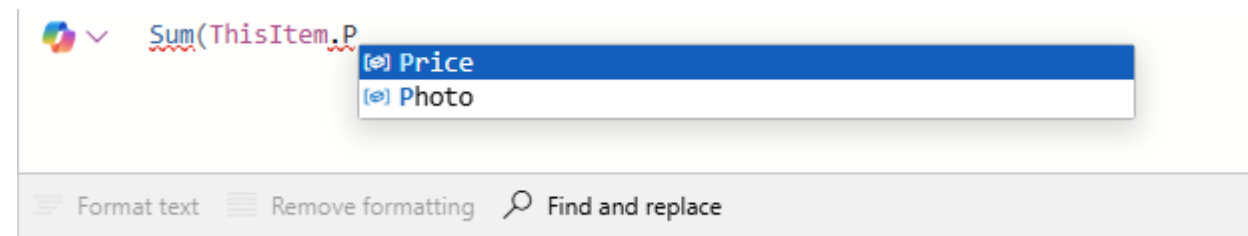
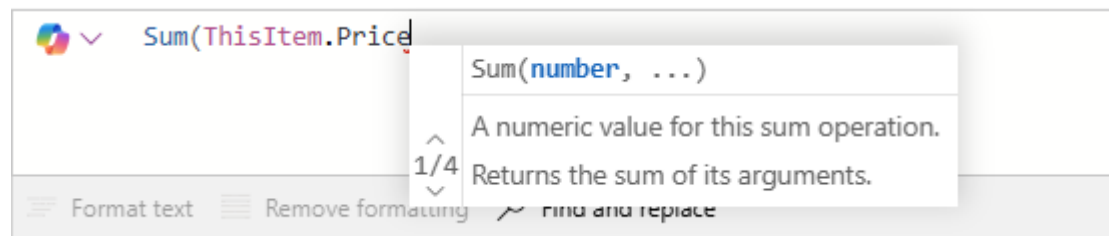
```
OrderBubbleTea( "milk tea", "50%", "half ice", 2 )  
OrderBubbleTea( "pizza", "100%", "less ice", "pearls" )  
OrderBubbleTea( "taro", "50%", "no ice", "extra ice", "pudding")  
OrderBubbleTea( "50%", "half ice")  
OrderBubbleTea( "Mala", "Friday", "less ice", "pearls" )
```

Function Error

PowerApps formulas – think Excel

And	Clock	DateDiff	Location	Notify	Set
App	Collect	Error	LookUp	Now	Sort
As	Color	Exit	Lower	Patch	Sum
Average	ColorValue	Filter	Max	Rand	Text
Back	Concatenate	Host	Mid	RandBetween	Today
Blank	Concurrent	If	Min	Refresh	User
Boolean	CountIf	IsMatch	Mod	Reset	Value
Calendar	CountRows	Launch	Navigate	Self	Year

Suggestions and Completion help



Types of Power Fx Formulas



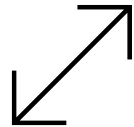
Functions

- Take parameters, perform an operation, return a value
- E.g. `Text(ThisItem.Date, "mm-dd-yyyy")`



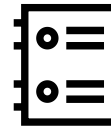
Signals

- No need to take parameters. Will give an output.
- E.g. `UTCNow()`



Enumerations

- Returns a pre-defined list of constant values (choices)
- E.g. `Color` is an enumeration that had pre-defined values such as `Color.Blue`, `Color.Black`, etc.



Named Operators

- Reserved keywords that are used by PowerApps
- E.g. `ThisItem` / `Parent`



Power Apps

Variables

Variables

- Variables temporarily hold data that can change dynamically
- Types include name, date/time, email address, etc.
- Like Excel formulas

	A	B
1	49	
2	65	
3	114	
4		

Text = fx TextInput1 + TextInput2

Screens

- Search
- App
 - Screen1
 - Label1
 - TextInput2
 - TextInput1

49

65

114

Data types

A data type is an attribute associated with a piece of data that tells a computer system how to interpret its value

- Number – 123.456
- Text – "Hello World"
- Boolean – True or False
- DateTime – 1/12/2017 12:00 PM
- Date – 1/12/2017
- All can hold blank (null) values

Why we need data types

Data types are like categories that help computers (and us) understand what kind of information we're working with. Just like in real life, where different items belong to specific categories, computers need to know what kind of data they're dealing with so they can handle it properly.

$$\begin{array}{|c|} \hline 1 + 1 = 2 \\ \hline \end{array}$$

$$\begin{array}{|c|} \hline "1" + "1" = 11 \\ \hline \end{array}$$


With the wrong data type, a **Text Concatenate operation** is performed instead of a **mathematical addition operation**

Item Name	Quantity Sold
Chicken Rice	Five
Nasi Lemak	Six

$$\begin{array}{|c|} \hline "Five" + "Six" = "FiveSix" \\ \hline \end{array}$$


Without numerical data types, instead of adding the quantity sold, a **Text Concatenate operation** is performed instead of a **mathematical summation**



Data types should be the **SAME** for Operations

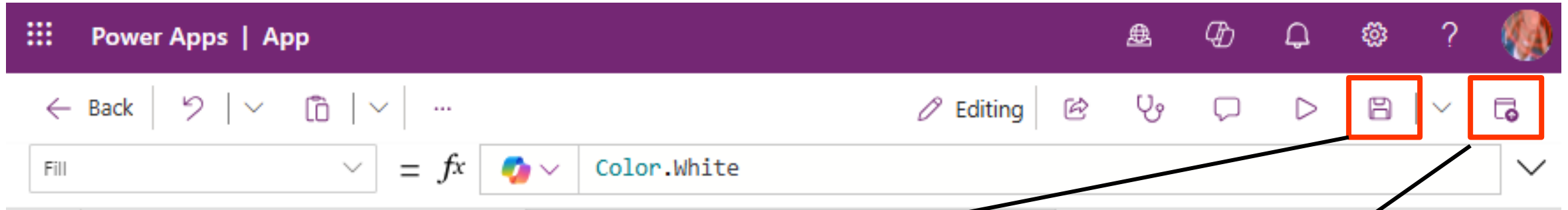
$$\begin{array}{|c|} \hline "Five" + 6 = \text{Error} \\ \hline \end{array}$$




Power Apps

Publishing an App

Saving vs Publishing



Saving – saves the new changes made to the app and increase the version number. This makes it available to all makers as the latest version.

Publishing – Releases the new version to all users so that they can see the new changes and use new features. E.g. New screens / buttons.

*** Save frequently, Publish only when you have fully tested an app and are confident the formulas and functionality works!

Publishing the App

- You may choose to change the following here:
- Name
- App icon
- Description

Publish

This version of your app will be published in the environment Reu-NexusSales.
[Learn more about publishing](#)

To improve app performance, app data may be locally stored in browser storage and some app assets may be loaded before users finish authentication.
[Learn more](#)

App icon and name

 App

Description

Provide a description to help end-users find and use your app.

Briefly describe what the app does

Publish this versionEdit details

Sharing the App

Power Apps | App

← Back | ↶ | ▾ | 📁 | ▾ | ...

Fill = fx Color.White

Editing | 📎 | 🔄 | 💬 | ▶ | 📁 | ▾ | 📄

What should it do?

List inventory | Manage inspections

Use everyday words to describe what your app should collect, track, list, or manage ...

This feature is in preview and uses generative AI [See terms](#)

Other ways to create an app

- Start with data**
Create new tables, select existing tables, or connect to external data sources.
- Start with a page design**
Select from a list of different designs and layouts to get your app going.
- Start with an app template**
Select from a list of fully-functional business app templates. Use as-is or customize to suit your needs.

Your apps

Name	Modified	Owner	Type
App	4 minutes ago	Reuel Ngo	Canvas
Event Attendance App		Reuel Ngo	Canvas
Sample1		Reuel Ngo	Canvas
Test		Reuel Ngo	Canvas

Context menu for 'App':

- Edit
- Play
- Share
- Details

Sub-menu for 'Share':

- Share
- Add to Teams

You can share the app via the share button here

Sharing the App

Specify whom you want to share the app with by name

Send them an email

Include an image in the email

Include email invitation checkbox

If you want the user to edit and share the app, select the co-owner checkbox
Best practice*

If you use certain data sources in your app, make sure your users have access to the data sources as well

Share App

Add people as Users and Co-owners to your app. Make sure your data connections have been shared with all users. [Learn more](#)

Enter a name, email address, or Everyone

New users

reuel1 ngo
User

Shared with

Sort by Name

Reuel Ngo
Owner

Deploy to a new environment
Deploying with a pipeline allows you to copy your app, and all objects associated with it, into a new environment. Consider using a pipeline to promote healthy application lifecycle management.

→ Start the deployment process

Email message
Let colleagues know what your app does and how it can help them.

Include an image
Add an image to the email to showcase what your app looks like.
Tip: Use an image that is 4:3 aspect ratio and smaller than 1MB.

Choose a file to upload or drag and drop it here.

Upload

reuel1 ngo
This user can use this app.

☐ Co-owner
Can use, edit, share app but not delete or change owner.

Data permissions ⓘ
Make sure your users have access to the data used in your app, including gateways, APIs, connectors, and tables.

Send an email invitation to new users

Share Cancel

Data permissions ⓘ

Make sure your users have access to the data used in your app, including gateways, APIs, connectors, and entities.

Excel

OneDrive for Business

Power Automate Fundamentals

Sing Kwang Eng

Cloud Solution Architect – Business Applications
Microsoft



What is Power Automate?

Time	Activity Title
0900hrs	Introduction to Power Automate
0930hrs	Exploring Power Automate Designer Studio, Copilot and common use cases
1015hrs	Tea Break
1100hrs	Hands on Power Automate Flow Labs – Part 1
1115hrs	Lunch
1245 hrs	Project Time

What is Power automate?

A comprehensive, end-to-end cloud automation platform powered by low code and AI that automates actions across the most common enterprise apps and services.

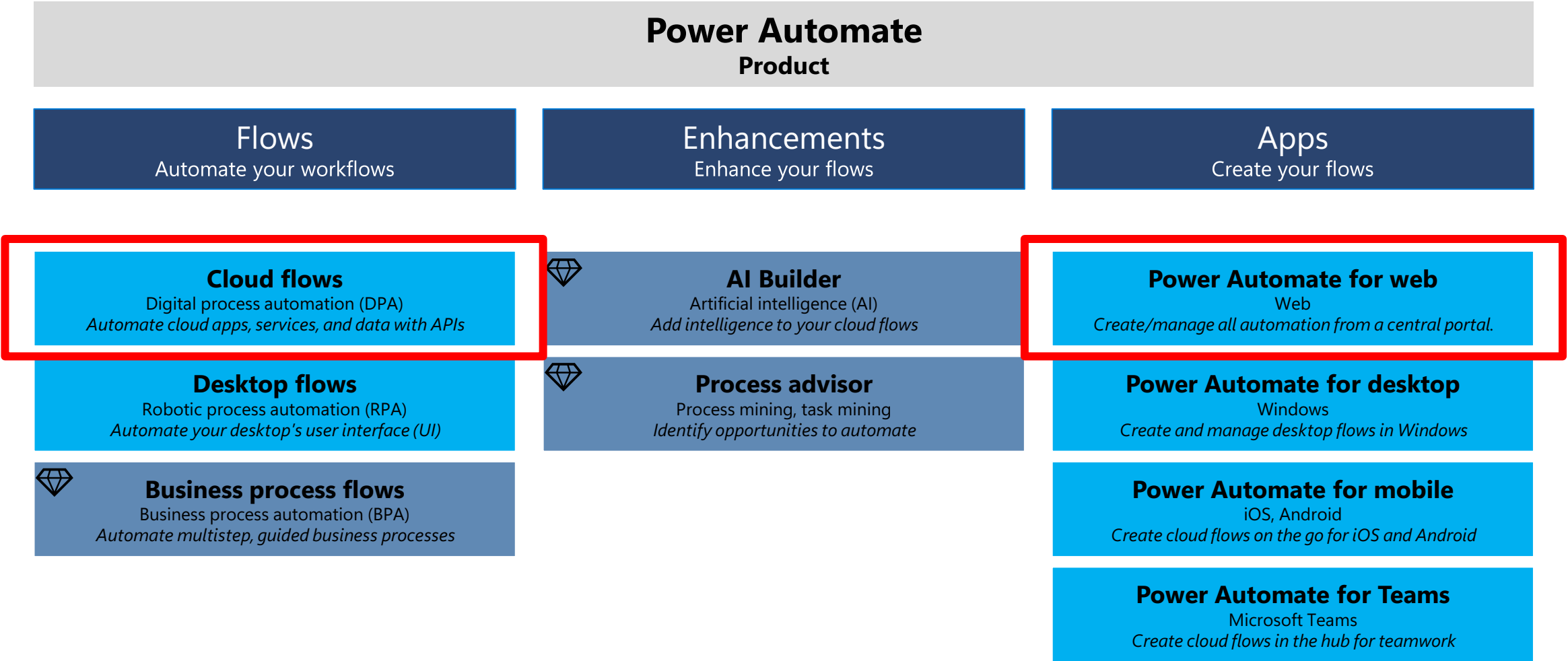
Basic Automation examples

- Post a message to an owner whenever a task is assigned to them in Microsoft planner
- Automatically save all email attachments from Outlook to OneDrive
- Generate and send weekly reports with SharePoint lists, excel
- Convert documents into pdf files when they are dropped into a folder
- Send emails from an excel list with user emails
- Getting approvals from manager

Advanced Automation examples

- Generate insights via GPT from scheduled email briefings
- Identify and notify users who create bad data in SharePoint lists
- Contract renewal reminders

Overview of Power Automate



Cloud Flows – Digital Process Automation

Industry Term

Digital process automation (DPA)

Description

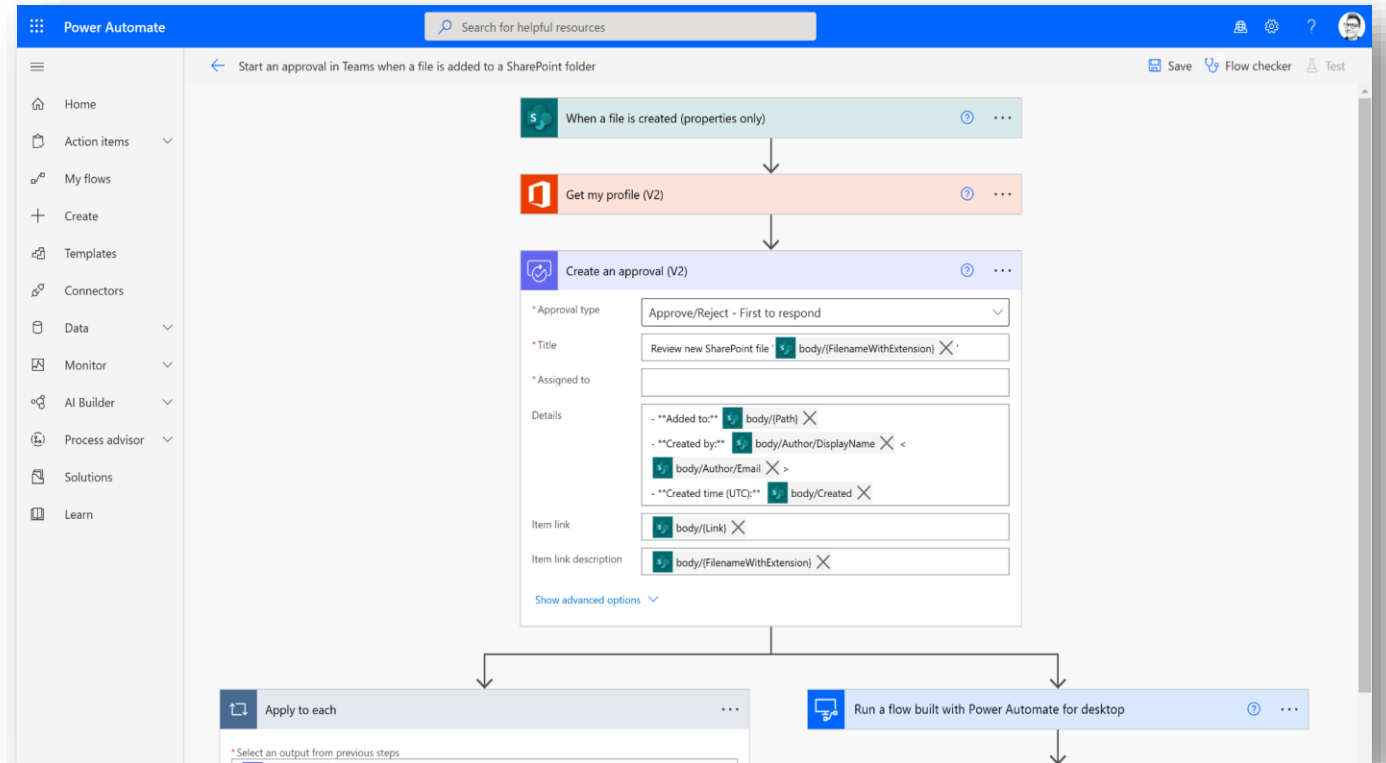
Automate between cloud-based apps, services, and data using API connectors and hundreds of prebuilt templates

Apps

Power Automate for web

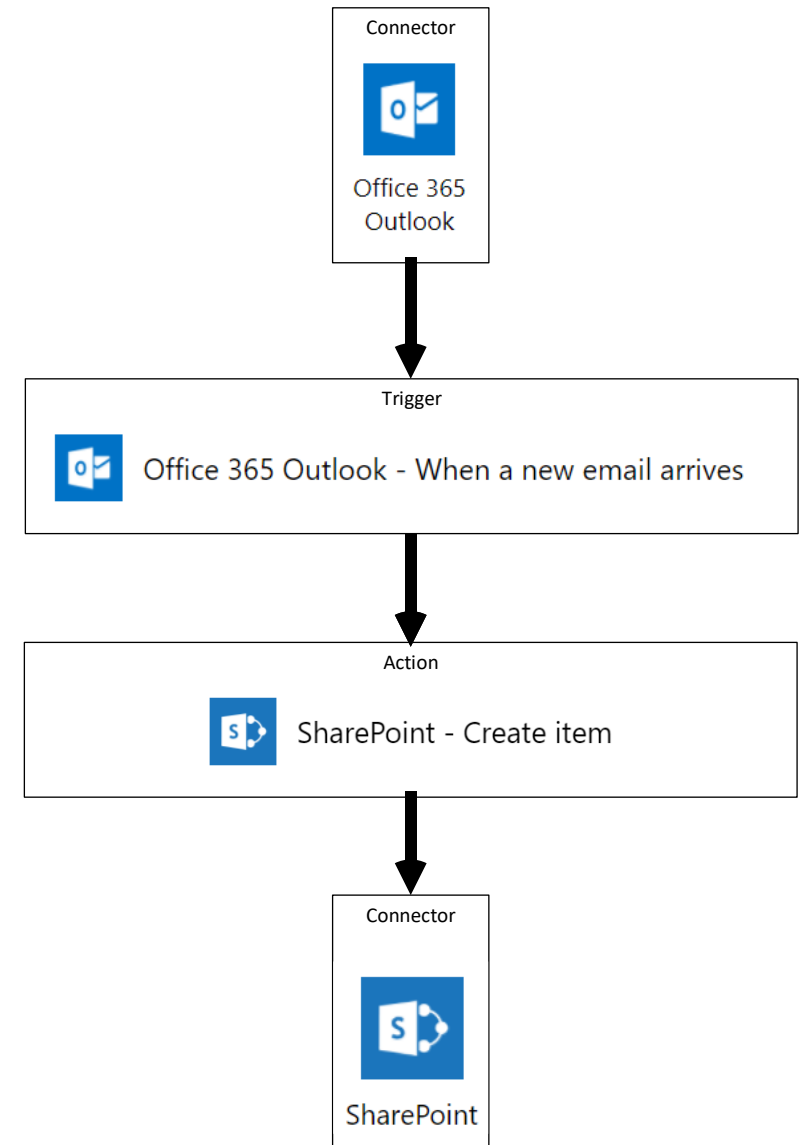
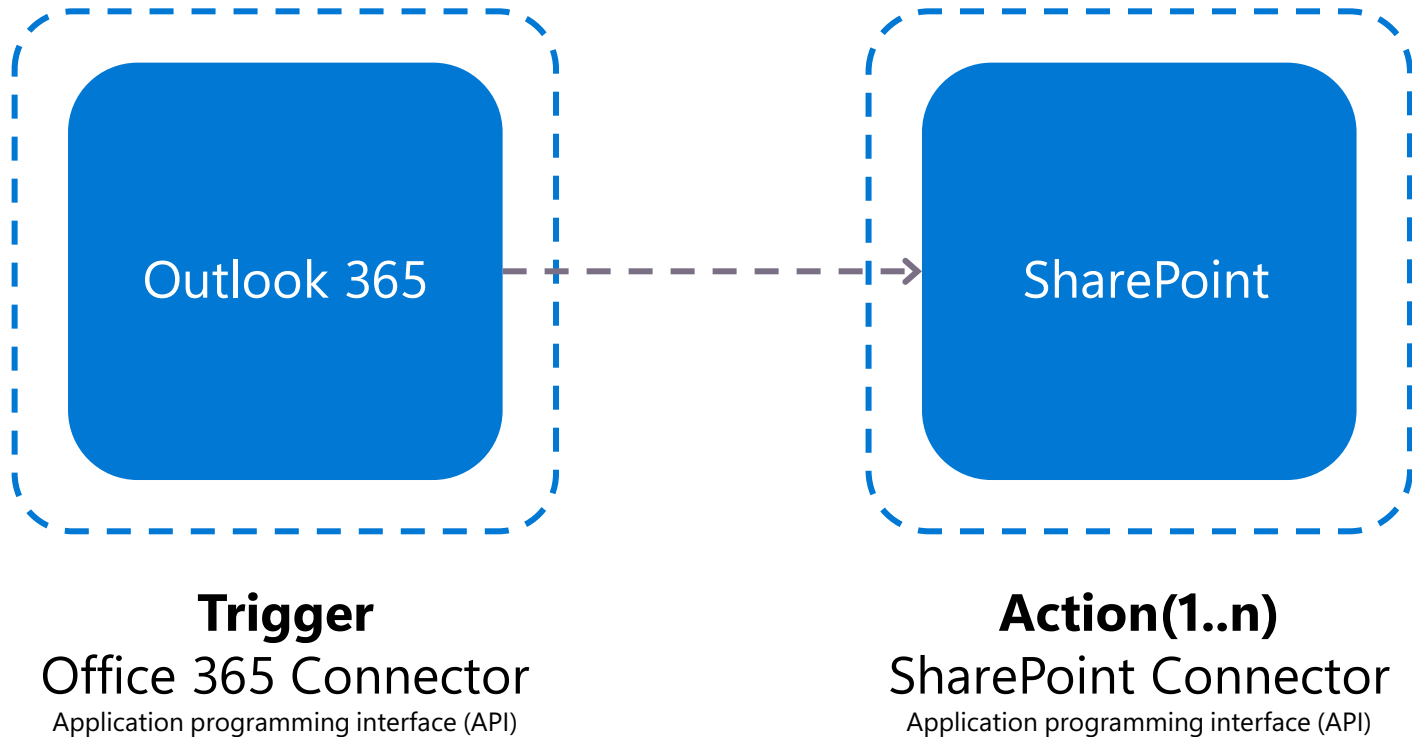
Power Automate for mobile

Power Automate for Microsoft Teams



How cloud flows work

Automate API cloud apps, services, and data using connectors



A closer look at cloud flows

Triggers

Choose how your cloud flows run

Automated

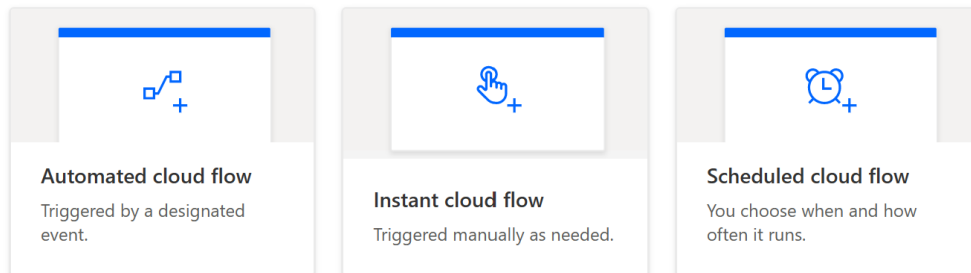
Run a cloud flow when something happens

Instant

Run a cloud flow manually

Scheduled

Run a cloud flow on a schedule



Connectors

Connect your cloud flows to apps, services, and data

Standard

Connect between a wide range of cloud apps, services, and data

Premium

Expand the cloud apps, services, and data you can connect to

Custom

Create custom connectors from scratch or build from an OpenAPI file or URL

[Connector reference overview](#) | [Microsoft Learn](#)



Azure AD



By: Microsoft



Azure AD App Registrations



By: Paul Culmsee and Microsoft



Azure App Service



By: Microsoft



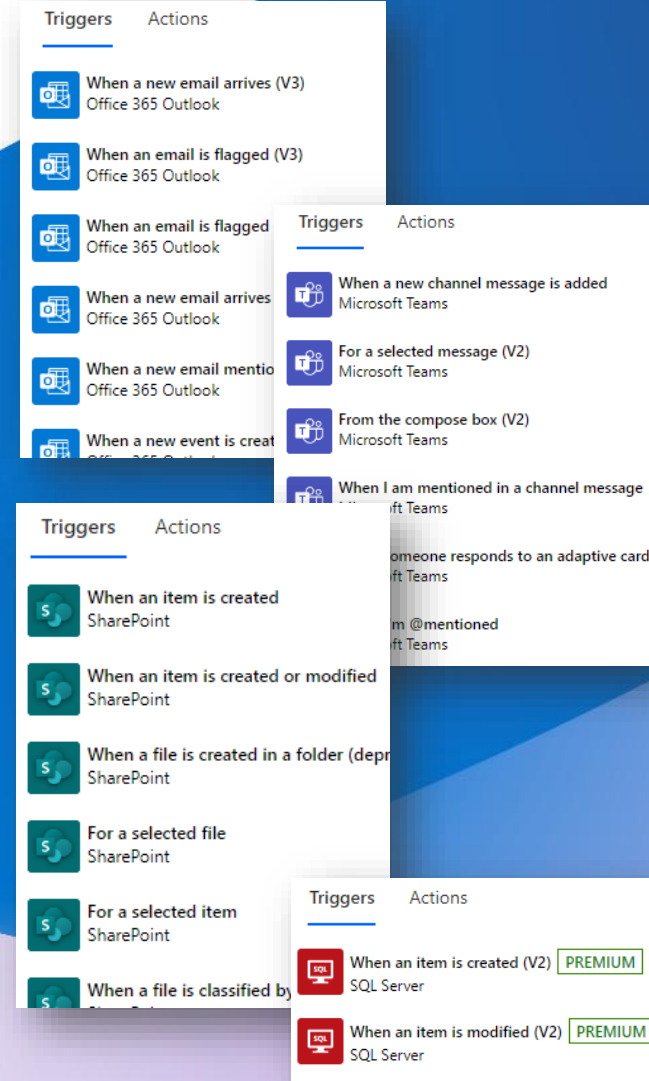
Azure Application Insights



By: Microsoft

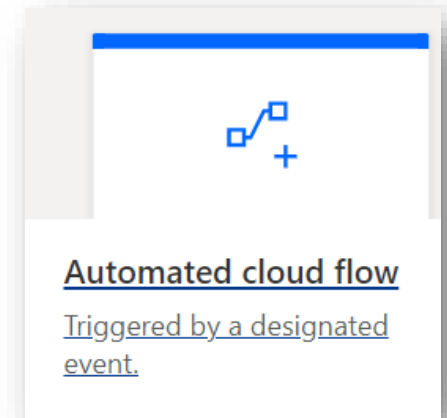
What is a trigger?

- A trigger is an event that starts a cloud flow. For example, if you want to get a notification in Microsoft Teams when someone sends you an email, in this case, receiving an email is the trigger that starts this flow.
- Triggers can be started instantly (manual), on a schedule, or automatically when an external event, such as when an email arrives
- Flow can have only one trigger



Automated cloud flows

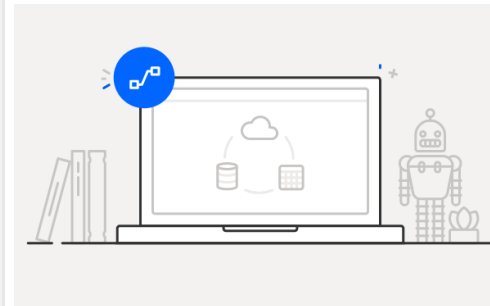
- Cloud flow that performs one or more tasks automatically after it's triggered by an event.
- For example, flow that notifies you by email when someone sends a tweet that contains a keyword you specify. In this example, sending a tweet is the event, and sending mail is the action.



Automated cloud flow

Triggered by a designated event.

Build an automated cloud flow



Free yourself from repetitive work just by connecting the apps you already use—automate alerts, reports, and other tasks.

Examples:

- Automatically collect and store data in business solutions
- Generate reports via custom queries on your SQL database

Flow name

Choose your flow's trigger * ⓘ

Search or select a trigger from the list below to create a flow. (Required)

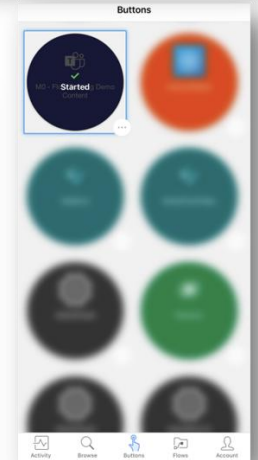
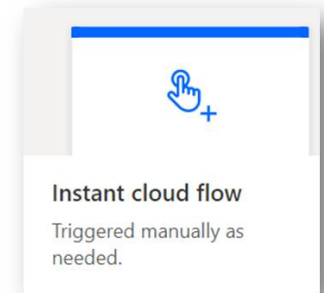
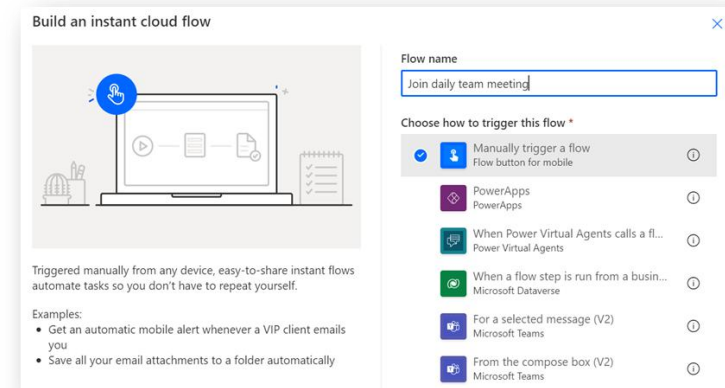


When a new tweet is posted
Twitter



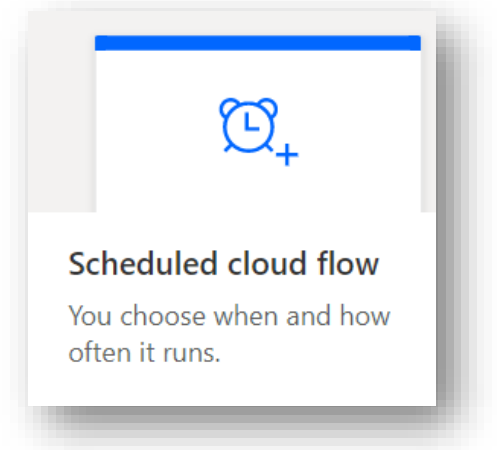
Instant (manual) cloud flows

- Cloud flow that performs one or more tasks after it's triggered manually
- For example, flow with a tap of a button on your mobile device, to remind your team to join the daily team meeting. You can trigger these flows manually from any device.
- Another instant flow example could be starting flow by clicking a button in Power Apps

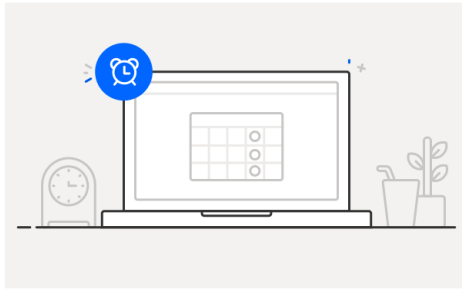


Scheduled cloud flows

- Cloud flow that performs one or more tasks after it's triggered by defined schedule like
 - Once a day, an hour, or a minute.
 - On a date that you specify.
 - After a number of days, hours, or minutes that you specify.
- For example, run a flow once per week to send a project report



Build a scheduled cloud flow



Stay on top of what's important without the effort—you choose when and how often the flow runs.

Examples:

- Automate team reminders to submit expense reports
- Auto-backup data to designated storage on a regular basis

Flow name

Send project report

Run this flow *

Starting 1/31/23 at 12:00 AM

Repeat every 1 Week

On these days

S M T W T F S

This flow will run:

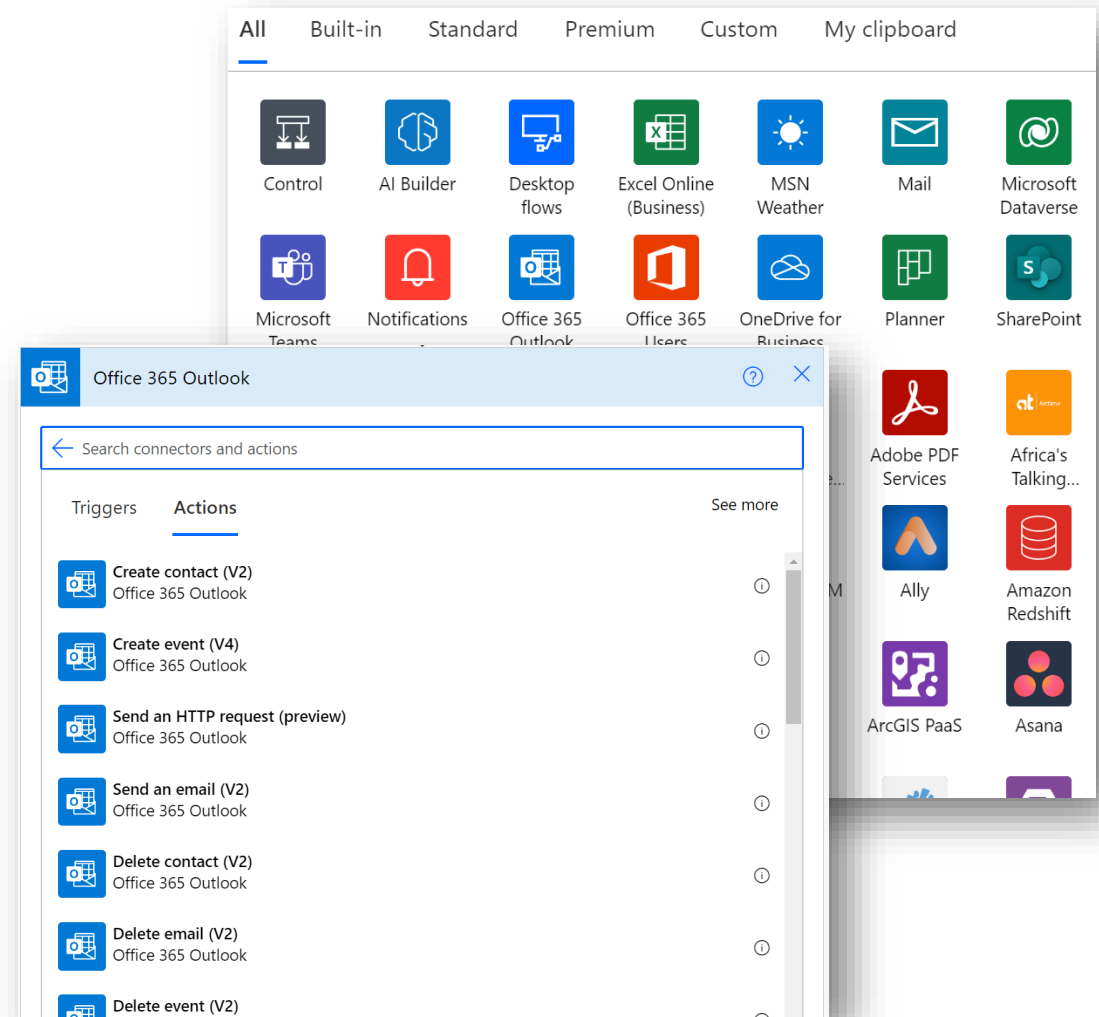
On Friday every week

What is an action?

Flow action defines an operation flow will execute in one step of the flow, for example, post a message to Team channel

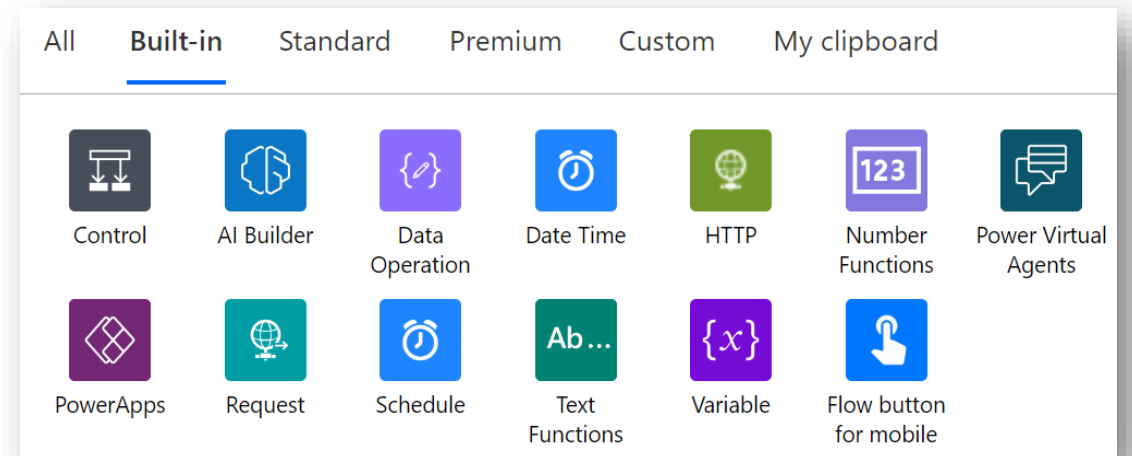
Flow can have multiple actions from any type of available connectors

Some actions offer advanced options like sends an email message as high priority or add attachments



Built-in Actions

- You can add more control and logic to the flow by using built-in actions
- Most common built-in actions are:
 - Control like Condition and Apply to Each
 - Data operations like Filter and Parse JSON
 - Variable actions



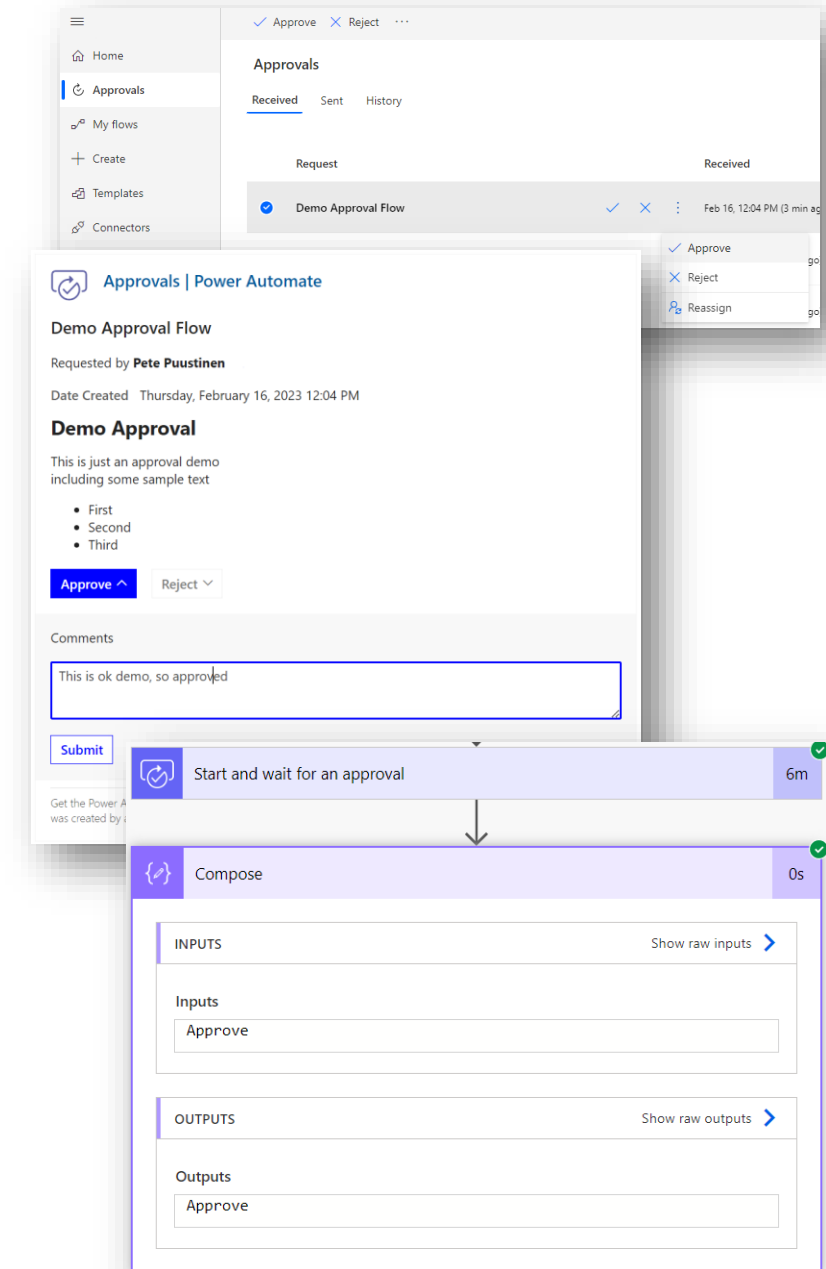
Power Automate Approvals

Whether you need written acknowledgment from your manager or a formal authorization from a diverse group of stakeholders, getting things approved is part of almost every organization.

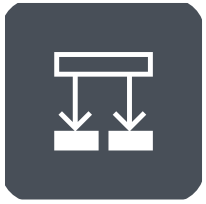
With the approvals capability in Power Automate, you can automate sign-off requests and combine human decision-making for workflows. Some popular cases where approvals can be used include:

- Approving vacation time requests.
- Approving documents that need sign-off.
- Approving expense reports.

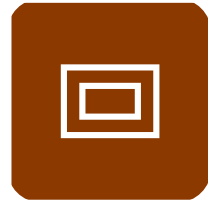
When you submit an approval in a flow, approvers are notified and can review and act on the request.



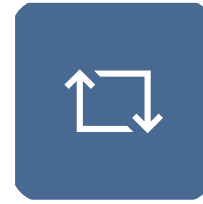
Advanced Actions



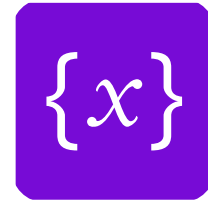
Condition



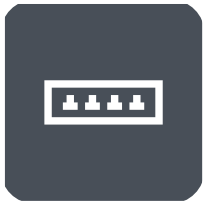
Scope



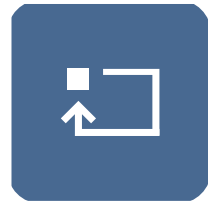
Apply to each



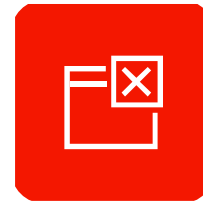
Variables



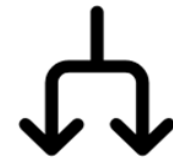
Switch



Do until



Terminate



Parallelism

Condition

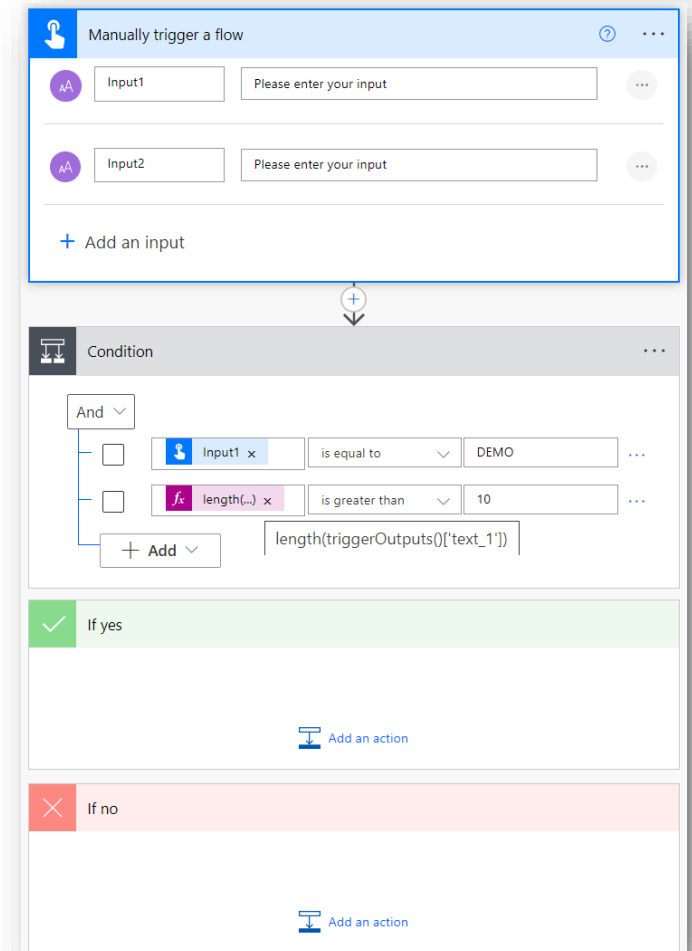
Use a condition to specify that a cloud flow performs one or more tasks only if a condition is true or false

Conditions include operators (basic mode):

- Contains / Does not contain
- Is equal to / Is not equal to
- Is greater than / Is greater than or equal to
- Is less than / Is less than or equal to

You can also use more than one expression, such as “If a parameter equals [X value or Y value]”

It is also possible to specify an expression in your conditions. This allows you to add more logic and control for conditions



Switch

Use switch case built-in action to have multiple conditions

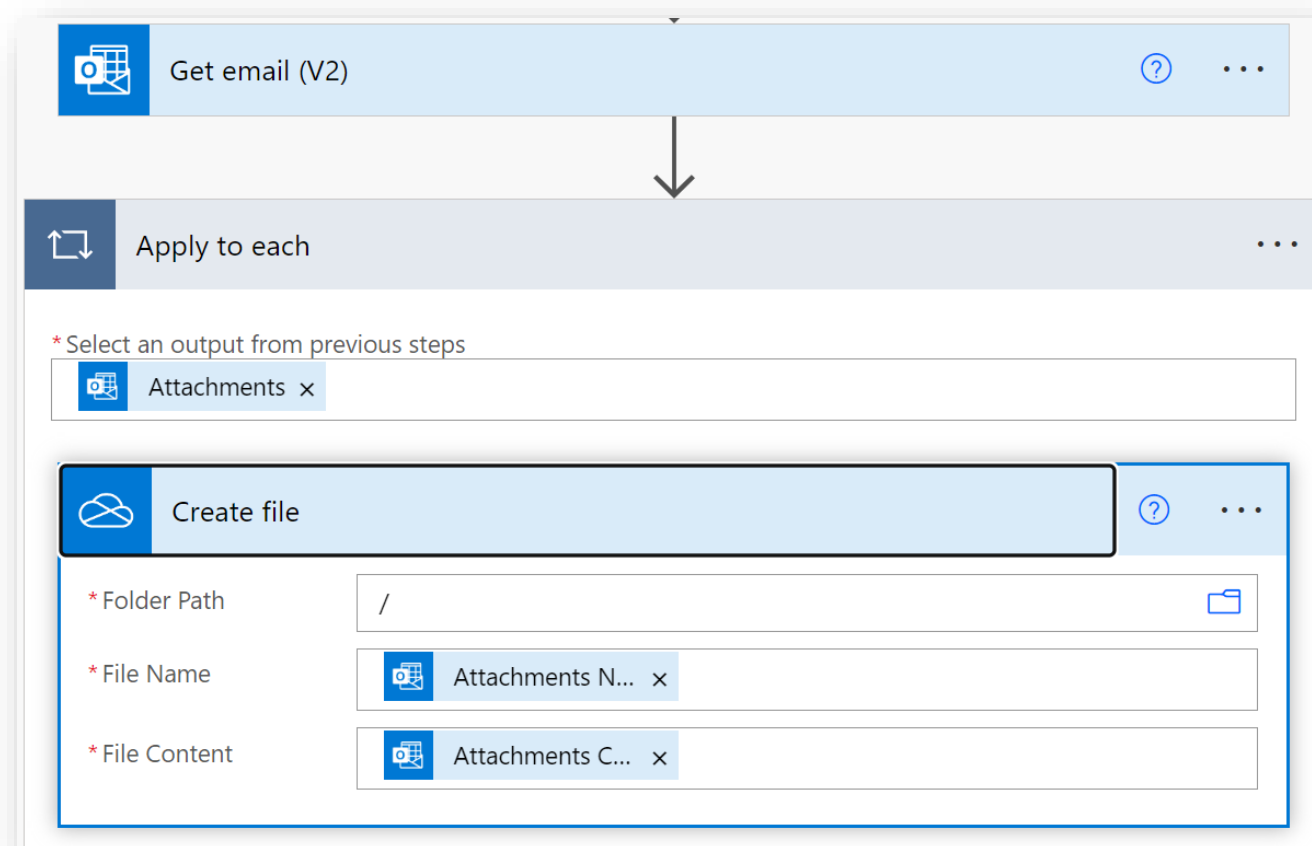
The screenshot displays a workflow editor interface for a 'Switch' action. At the top, a 'Switch' header is followed by a configuration box where the condition is set to '* On' and the field is 'Subject'. Below this, there are three distinct cases:

- Case**: The condition is '* Equals' with the field 'Approver'. The associated action is 'Send Approval Notification'.
- Case 2**: The condition is '* Equals' with the field 'Requestor'. The associated action is 'Send Confirmation'.
- Default**: A section for 'If no case contains a matching value' containing the action 'Notify Supervisor'.

Each case and the default section include an 'Add an action' button at the bottom. The interface uses a clean, modern design with light blue accents for action buttons and a clear hierarchy for the switch cases.

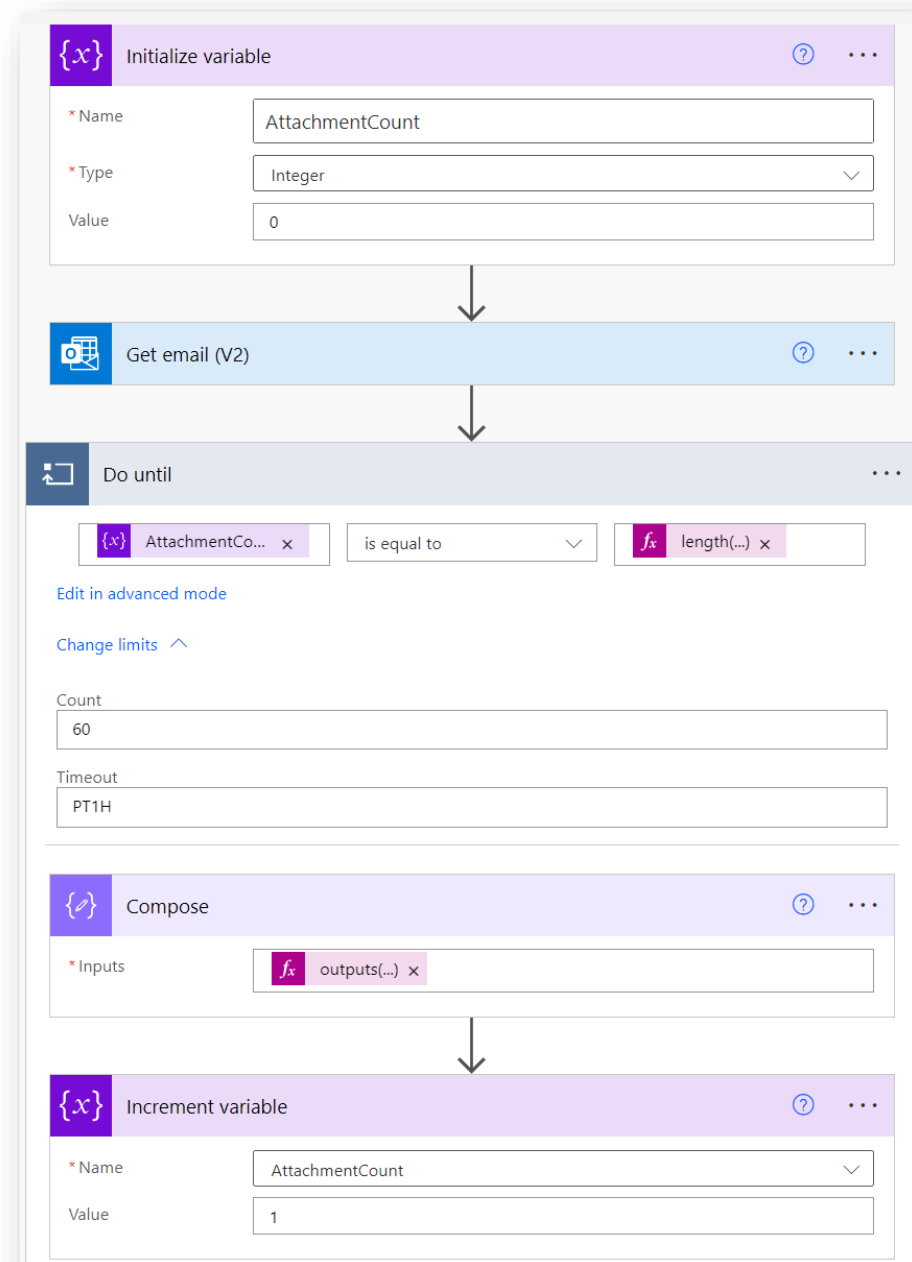
Apply to Each

Use Apply to each loop to process a list of items. For example, you could use apply to each to save email attachments to OneDrive



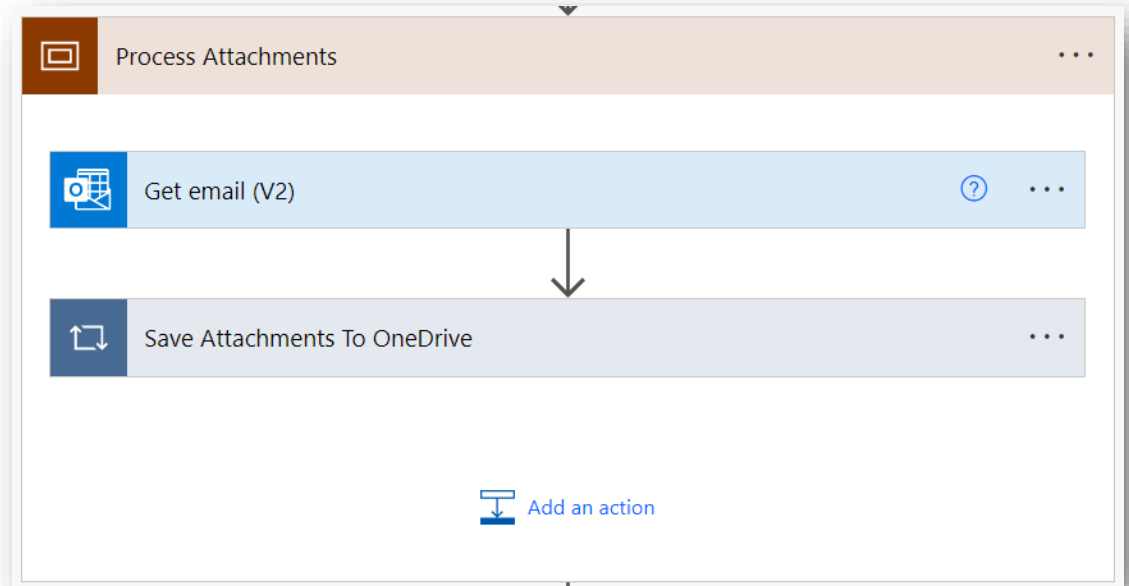
Do Until

- Use Do Until loop to run actions in the loop as many times as the defined condition
- If the condition is met, then loop exits i.e. it will run in the loop as long as condition is not true
- You can change the maximum count (5000) and timeout limits



Scope

- Use scopes to group actions and keep you flow better organized
- Also easier to move multiple actions around and even copy-paste
- Used also in the error handling (discussed in more detail later)



Variables

Variables can be used to store values in your cloud flows.

For example, track the number of times a loop runs. To iterate over an array or check an array for a specific item, you can use a variable to reference the index number 'apply to each' array item.

Supported datatypes are:

- Boolean
- Integer
- Float
- String
- Object
- Array



Append to array variable
Variable



Append to string variable
Variable



Decrement variable
Variable



Increment variable
Variable

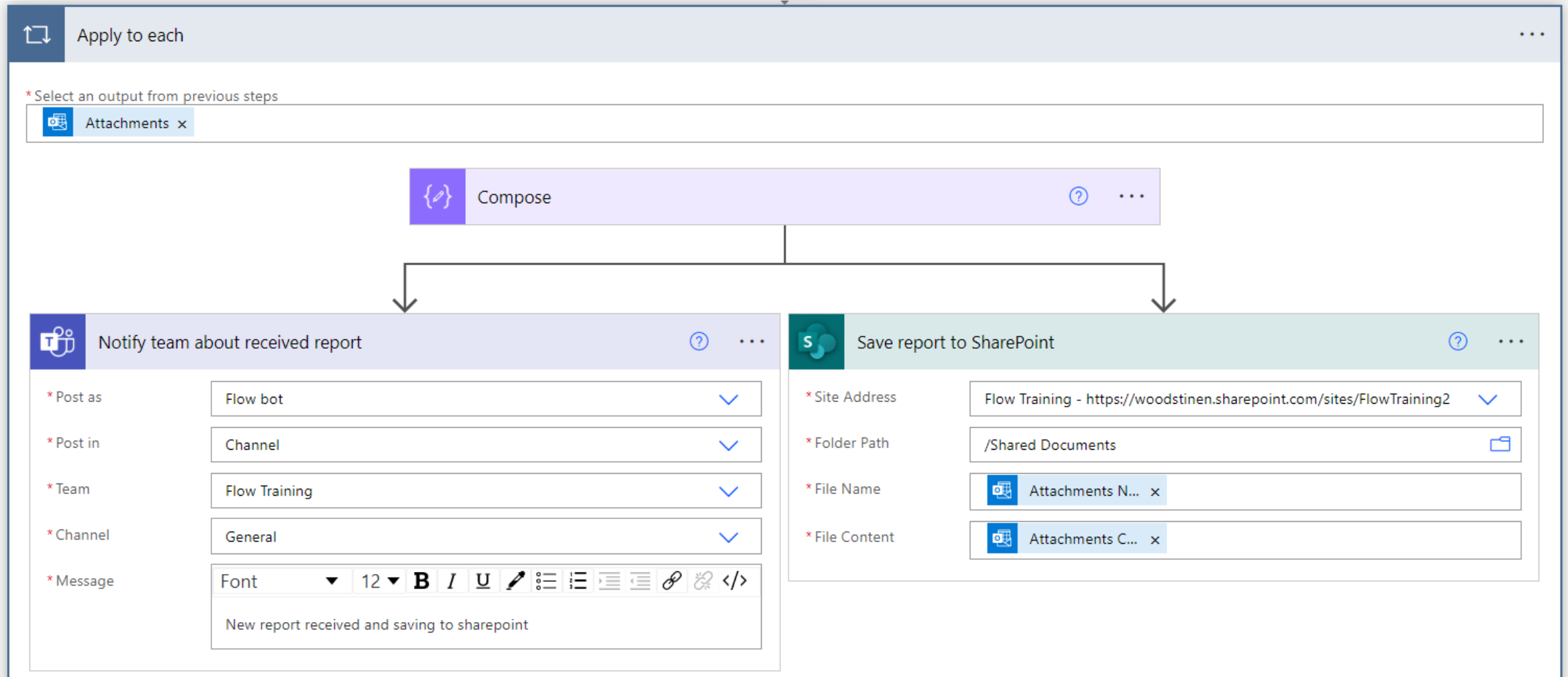


Initialize variable
Variable



Set variable
Variable

Run in Parallel - two or more steps that run at the same time





Connectors

Popular Standard connectors



Excel Online
(OneDrive)



Office 365
Outlook



OneDrive for
Business



Microsoft Teams



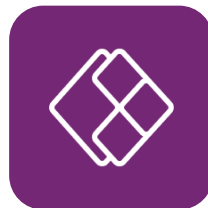
Microsoft To-Do
(Business)



Office 365
Users



Microsoft Forms



Power Apps
Notification V2



SharePoint

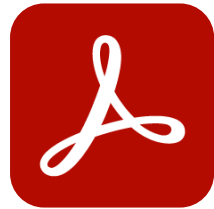


Planner

Popular Premium connectors



Dataverse



Adobe PDF
Services



SQL Server



DocuSign



Azure OpenAI



Calendly Forms



ClickSend SMS



ServiceNow

Independent Publishers

- Connectors that are developed, tested, published, and supported by connector developers that are independent from the end service or API.
- Independent Publishers include Microsoft employees, MVPs, developers, and other community members.
- Independent publishers have to go through a certification process for the connectors they wish to publish.

They can be free but may be blocked by your organization.



OpenQR

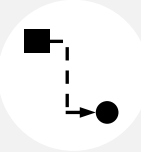


IQAir



Your First Flows

How to start using Power Automate

	Start with...	Progress to...	Automation Team...
Target users	 Personal	 Team	 Company
Complexity	 Low	 Medium	 High
What to automate?	 Partial Workflow	 Single Workflow	 Multiple Workflows

Example: Translating processes to Cloud Flows

Invoice Document Process

A building management uses vendors to perform services at their buildings, such as cleaning properties after tenants leave and providing landscaping services at their properties. After the jobs are completed, vendors send invoices to the building management company. The building management company might want to automate processing invoices from vendors.

Currently the process is as follows:

1. Invoices are received as email attachments. The management company uses Microsoft Exchange for processing emails.
2. These attachments are downloaded and stored in Microsoft SharePoint.
3. Each invoice is sent to a specific person for approval. Approval is made in Microsoft Teams.
4. Once approved, the invoices are entered into the company's ERP system. The management company uses Oracle. Once entered, a confirmation email with the invoice number must be sent back to the vendor.

Example: Translating processes to Cloud Flows

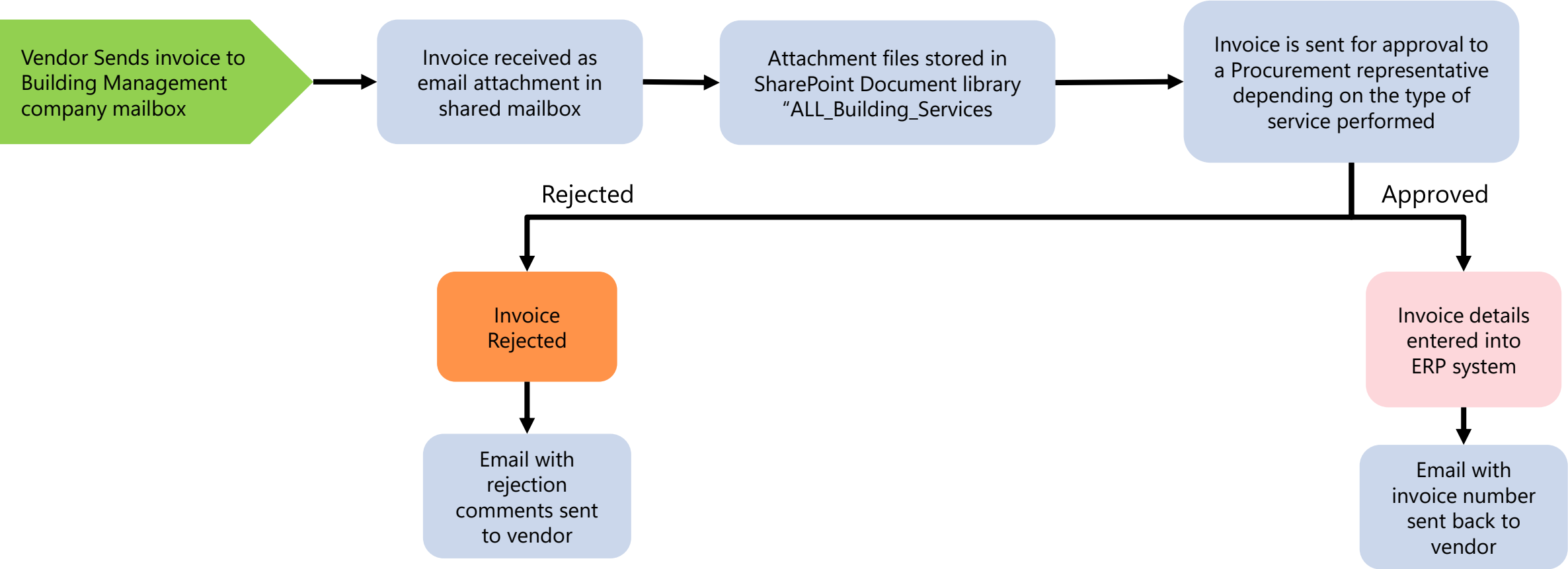
Invoice Document Process

Automating a process like this requires interaction with multiple services. There are likely different accounts being used with each service, and different data operation that are being executed. In this example, we would be working with at least four different services.

- **Office 365 Outlook:** First, you would need to monitor a specific Microsoft Exchange mailbox.
- **Microsoft SharePoint:** Attachments are downloaded and saved to a specific folder in Microsoft SharePoint.
- **Microsoft Teams:** Invoice approval requests are sent to managers in Microsoft Teams, where they can either approve or deny the request.
- **Oracle:** Once approved, the new invoice is created in Oracle. Details of the invoice are stored so they can be used later.
- **Office 365 Outlook:** A confirmation email is sent using a dedicated mailbox to the company that submitted the invoice.

Example: Translating processes to Cloud Flows

Invoice Document Process Mapping



Copilot in power automate

The background of the slide is a gradient of blue, transitioning from a darker shade at the top to a lighter shade at the bottom. On the right side, there are several curved, glowing white lines that sweep across the frame. These lines are composed of many small, bright white dots, giving them a sense of motion and energy. The overall aesthetic is modern and technological.

Transform natural language to Automation

Bring ideas to life with AI powered automation development

Now you can **describe what you want** to automate in a sentence, and an AI-based copilot will build your flow in seconds

This new way to create flows runs on OpenAI Codex, an AI model descendant of GPT-3 that can **translate natural language to code**, in this case, Power Automate cloud flows

The screenshot shows the 'Describe it to design it (preview)' interface in Power Automate. The interface is divided into two main sections. On the left, under 'Step 1 of 3', is the 'Start building a cloud flow with your own words' section. It includes a sub-header 'Just describe what you want to automate and built-in AI assistance will help you create your flow. [How it works](#)'. Below this, it states 'This preview feature works with many common actions in the following Microsoft 365 apps, with support for more actions and connections coming soon:'. A list of supported apps is shown with icons: Approvals, Dataverse, Excel, Forms, Microsoft Teams, Office 365 Outlook, OneDrive for Business, Planner, and SharePoint. On the right is the 'Describe your flow in everyday words' section. It features a text input field containing the sentence 'Send me a message on Teams every time a new file is added to SharePoint' with a right-pointing arrow button. Below this input field is a section titled 'Or try one of these examples to get started' containing three example flows: 'Create a task in planner when someone responds to a Microsoft Forms survey', 'Send an automatic message to a Teams channel once every week', and 'When someone mentions me by email, send me a message on Teams'. A 'View more examples' link is at the bottom right of this section. At the bottom of the interface, there are 'Next' and 'Cancel' buttons.

Create, edit and explain with copilot

The screenshot displays the Power Automate web interface. The top navigation bar is blue and contains the 'Power Automate' logo, a search bar, and the 'Environments DemoEnv' dropdown. Below the navigation bar, a toolbar includes links for 'Send feedback', 'Copilot', 'Save', 'Flow checker', 'Test', and 'New designer'. The main workspace is a large grid with a single action box labeled 'Manually trigger a flow'. On the right side, the 'Copilot' sidebar is open, showing a welcome message and instructions on how to use the AI assistant to create or modify flows.

Power Automate

Search

Environments DemoEnv

← Dummyapproval

Send feedback Copilot Save Flow checker Test New designer

Manually trigger a flow

+

+

-

↻

🔍

📌

Copilot

Welcome to Copilot in Power Automate

Be more efficient than ever with AI assistance. Simply tell Copilot what you want to do and it will help you get started.

1 of 3 [Next](#)

Welcome back! If you want me to change your flow, just say what you want. For example:

- Add an action that sends an email
- Explain what an action does
- Add a condition

[Test this flow](#)

Ask a question or describe how you want to change this flow

0/2000

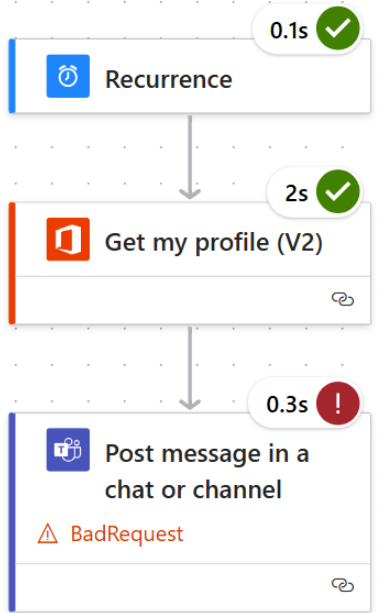
Make sure AI-generated content is accurate and appropriate before using.

Now you can **describe what you want** to automate in a sentence, and an AI-based copilot will build your flow in seconds

This new way to create flows runs on OpenAI Codex, an AI model descendant of GPT-3 that can **translate natural language to code**, in this case, Power Automate cloud flows

Troubleshoot with copilot

Identify and resolve errors that might occur during the testing of cloud flows



0.1s ✓

Recurrence

2s ✓

Get my profile (V2)

0.3s !

Post message in a chat or channel

BadRequest

Copilot

To send a message in Teams using Power Automate, you can use the "Post a message in a chat or channel" action. You can send the message as the Flow bot or as a user. You can specify the team, channel, or group chat where you want to send the message. You can also mention a user in the message using the "Get @mention token for a user" action. Additionally, you can create a new group chat and post a message to it using the "Create a chat" action.

AI-generated content may be incorrect

I found the following error:

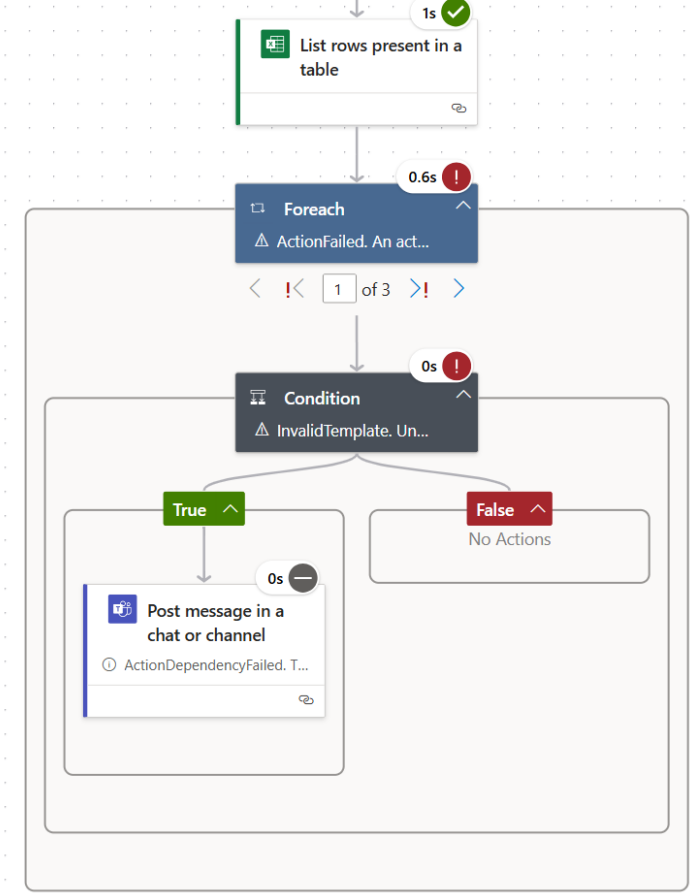
Post message in a chat or channel teams

In your 'Post message in a chat or channel' operation, the 'recipient' parameter is set to 'Dimitri', but no user details with that email or UPN were found in Graph. Double-check the email or UPN you are using and ensure it is correct.

Edit flow

AI-generated content may be incorrect

Make sure AI-generated content is accurate and appropriate before using.



1s ✓

List rows present in a table

0.6s !

Foreach

ActionFailed. An act...

1 of 3

0s !

Condition

InvalidTemplate. Un...

True

False

No Actions

0s -

Post message in a chat or channel

ActionDependencyFailed. T...

<insert certification name> has expired. For renewal, please contact <insert generic response>

Updated this action:

Post message in a chat or channel Microsoft Teams

Check these actions to see if any parameters need to be set.

Undo

AI-generated content may be incorrect

I found the following error:

Condition

The error occurred in the 'Condition' action. The error message indicates that the 'addDays' function expects its first parameter to be a string that contains the time, but it is receiving a null value. To fix this error, you should check the previous operation outputs and ensure that the parameter being passed to the 'addDays' function is a valid string. Double-check that you are using the correct parameter name and that it is not null or empty.

Edit flow

AI-generated content may be incorrect

Make sure AI-generated content is accurate and appropriate before using.

Thank you

Sing Kwang Eng

v-kwasing@microsoft.com



Please help to fill up feedback survey
with below URL or QR code above

<https://forms.office.com/r/XMnvZ0jdmZ>